ITC 215 – Introduction to Programming with Java

Assignment 2: Loops and Nested Loops

Instructor: Dr. Sami Naji

Name: Sewita Ahmadi

Due date: October 12,2026

Task 1.1

```
int total = 0;
for (int i = 1; i <= 4; i++) {
    total = total + i * 2;
    System.out.println("i = " + i + ", total = " + total);
}
```

Trace Table

Pass	i	Total(after)	Printed line
1	1	2	i = 1 , total = 2
2	2	6	i = 2 , total = 6
3	3	12	i = 3 , total = 12
4	4	20	i = 4 , total = 20

Exact output

i = 1, total = 2

i = 2, total = 6

i = 3, total = 12

i = 4, total = 20

Guiding Questions

1. What is the value of total before the loop starts?

total = 0.

2. How many times does the loop body run? How do you know?

Four times, because i takes the values 1, 2, 3, and 4, and the condition is i <= 4.

3. After the last pass, i becomes 5. Why does the loop stop?

The condition i <= 4 becomes false when i = 5.

4. Now run the code. Does your trace table match the real output?

Yes, the trace table matches the output shown above.

Task2.2

```
int n = 20;
int steps = 0;
while (n > 3) {
```

```
    n = n - 6;
    steps++;
}
System.out.println("n = " + n + ", steps = " + steps);
```

Trace Table

Pass	n (after)	steps (after)
1	14	1
2	8	2
3	2	3

Exact output

n = 2, steps = 3

Guiding Questions

1. Write n and steps after each pass of the loop in a small table.

Pass 1: n = 14, steps = 1

Pass 2: n = 8, steps = 2

Pass 3: n = 2, steps = 3

2. At what value of n does the condition n > 3 become false?

It becomes false when n = 2.

3. What is the final line printed? Run the code to check.

n = 2, steps = 3

4. What would happen if you removed the line n = n - 6;? Explain in one sentence.

The loop would run forever because n would stay at 20, so n > 3 would always be true.

Task 4.1

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= 2; j++) {
        System.out.println("i = " + i + ", j = " + j);
    }
}
```

Exact output

i = 1, j = 1

i = 1, j = 2

i = 2, j = 1

i = 2, j = 2

i = 3, j = 1

i = 3, j = 2

Guiding Questions

1. For i = 1, what values does j take? What happens to j when i becomes 2?

When i = 1, j takes the values 1 and 2. When i becomes 2, the inner loop starts again, and j resets to 1.

2. Is there a relationship between the number of outer passes, inner passes, and total prints?

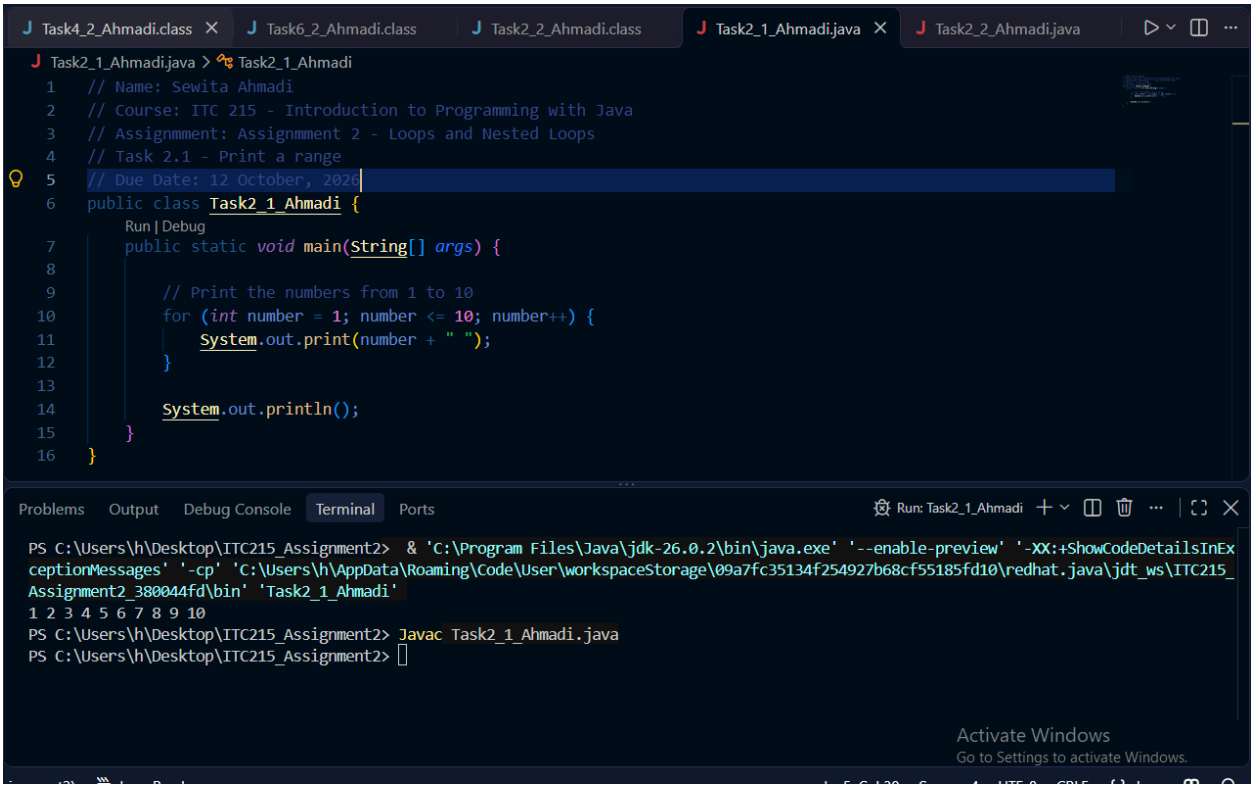
Yes. There are 3 outer passes and 2 prints per outer pass, so there are printed lines.

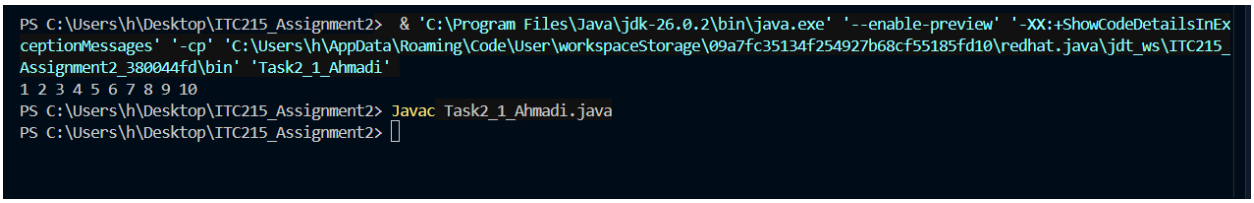
3. What would the total be if the loops were i <= 3 and j <= 4?

There would be printed lines.

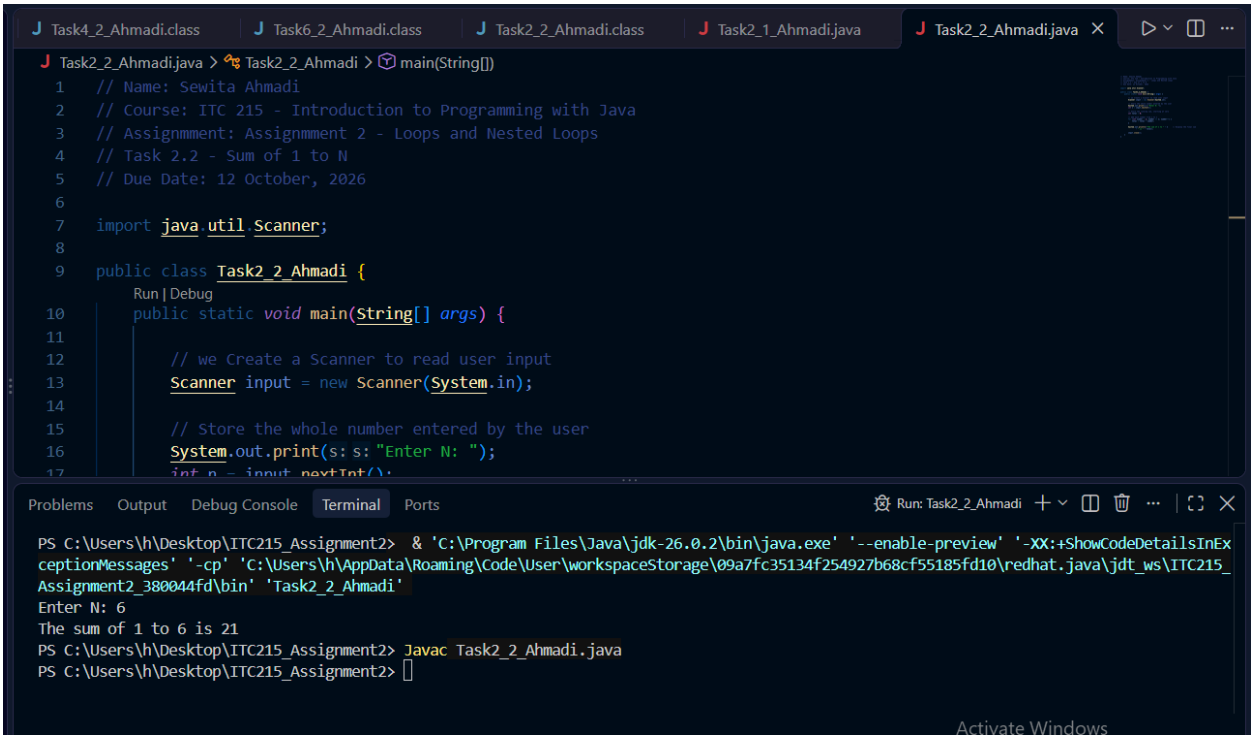
Part 2: Basic Loops

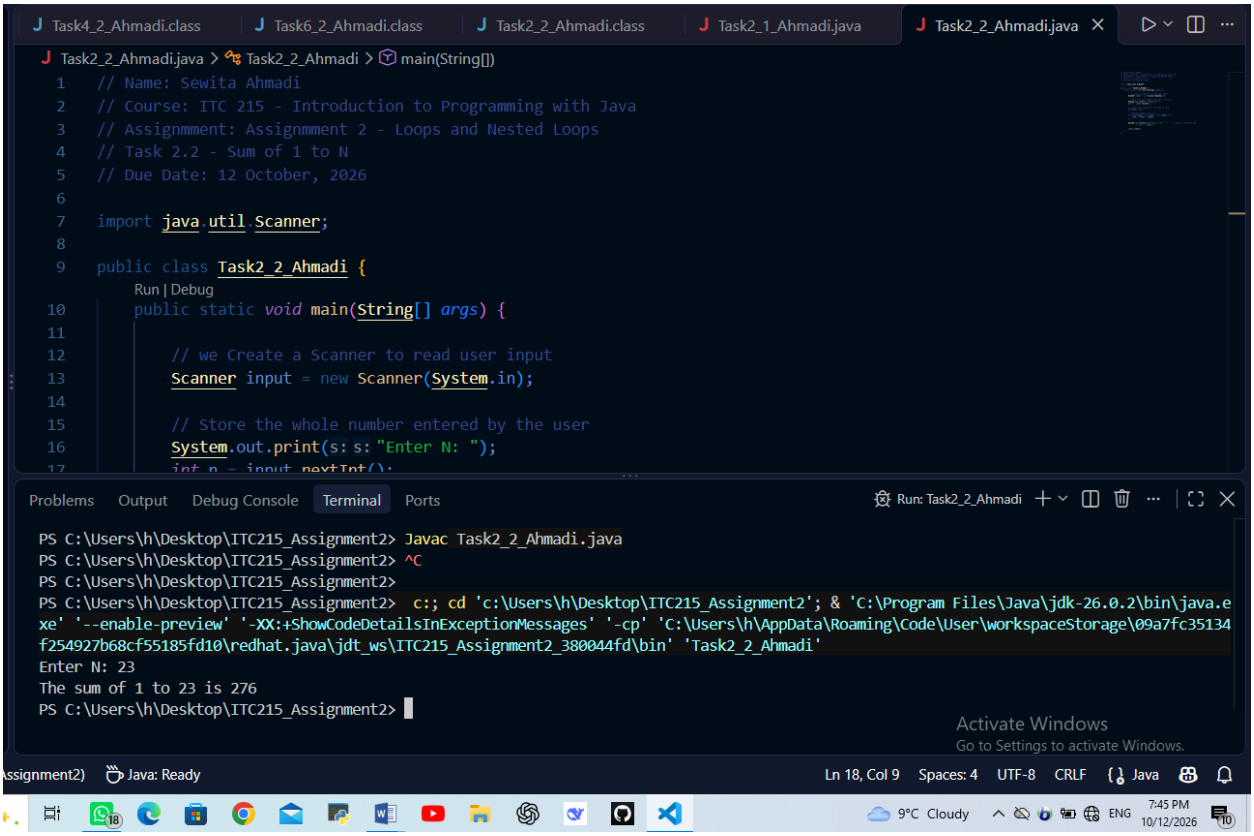
Task 2.1 Output Screenshot



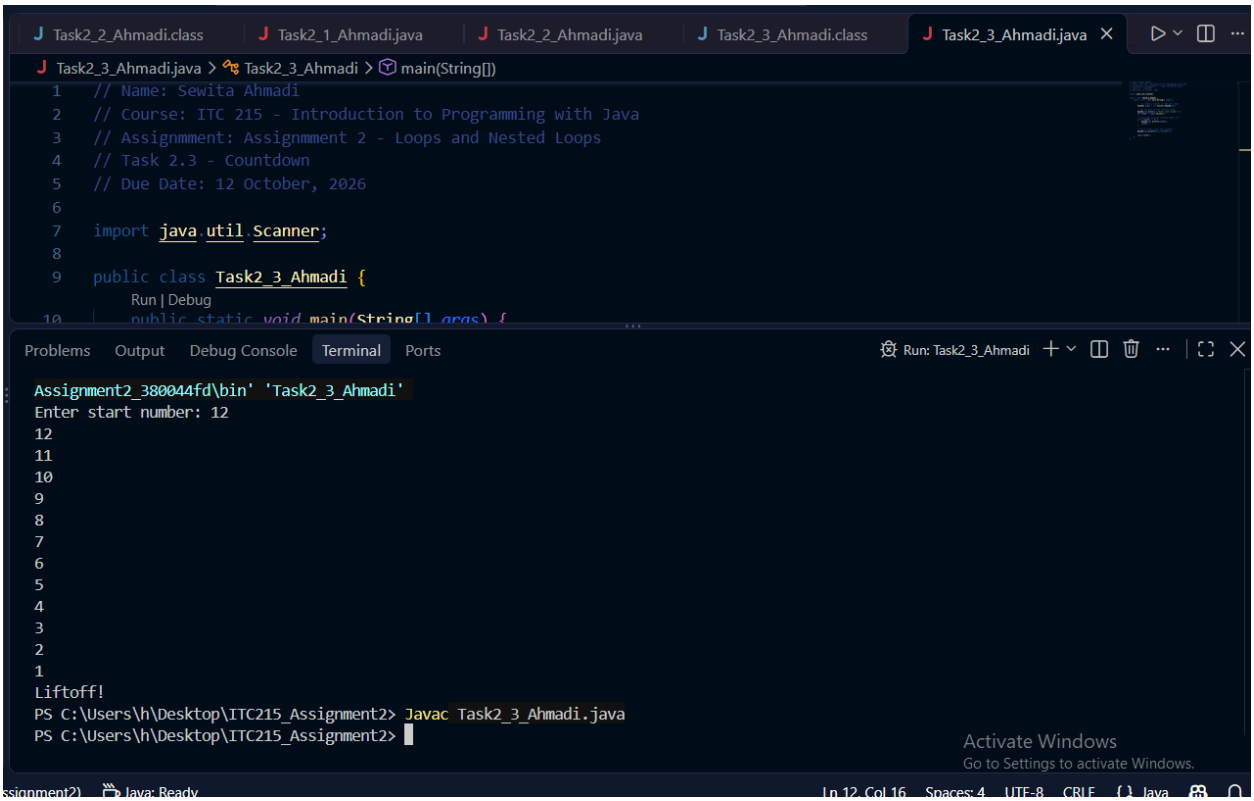


Task 2.2 Output Screenshots



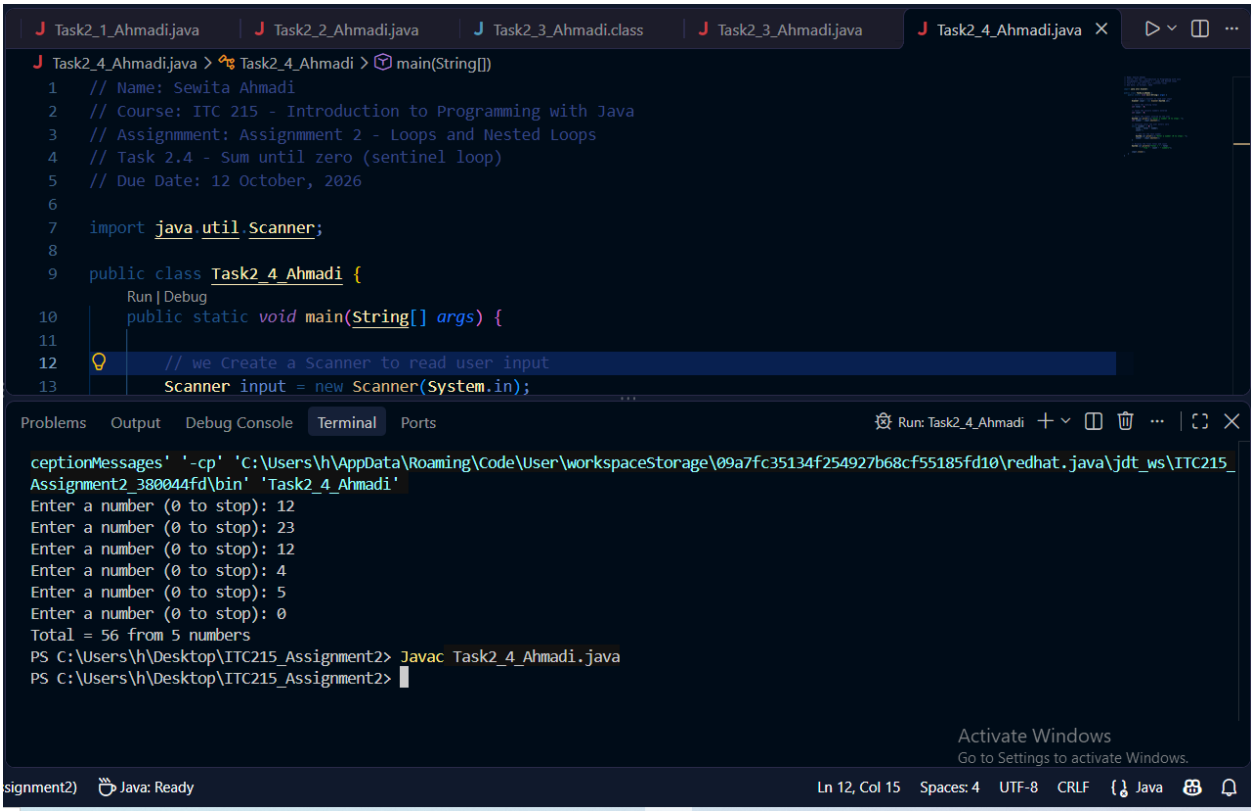


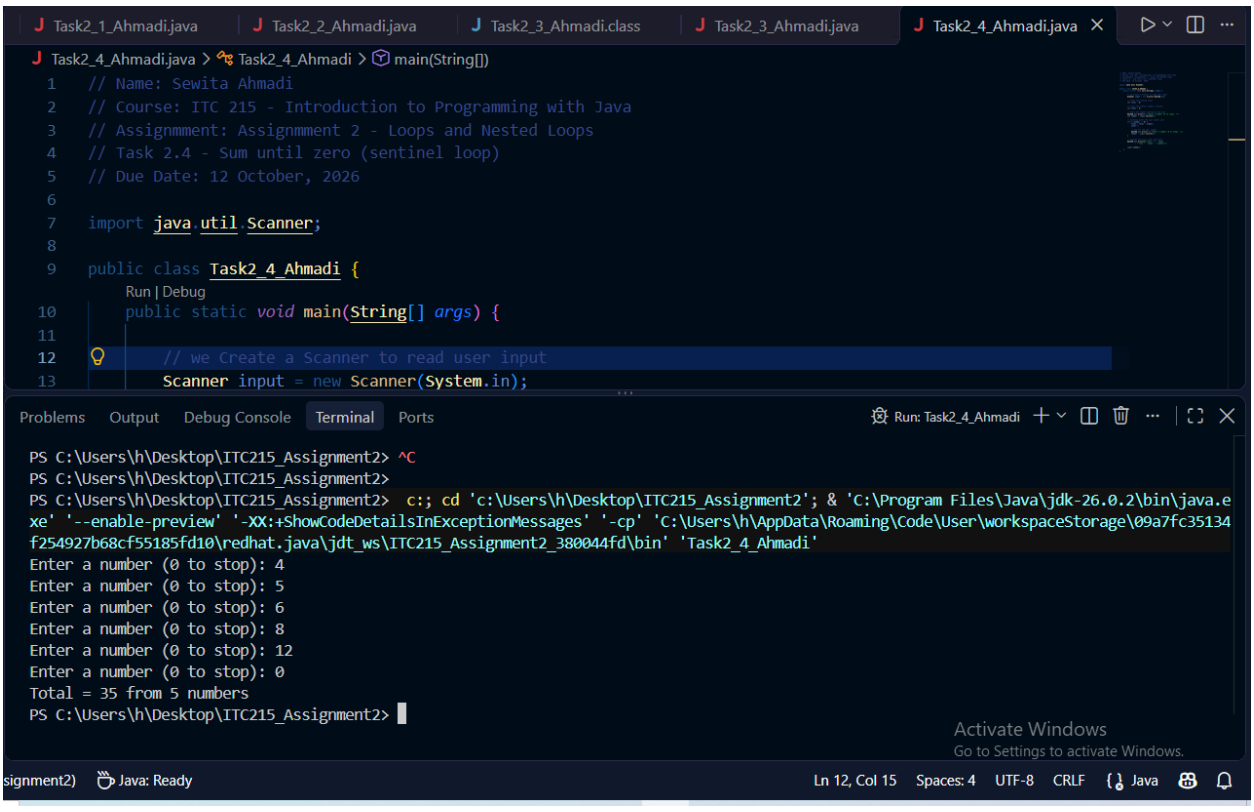
Task 2.3 Output Screenshot



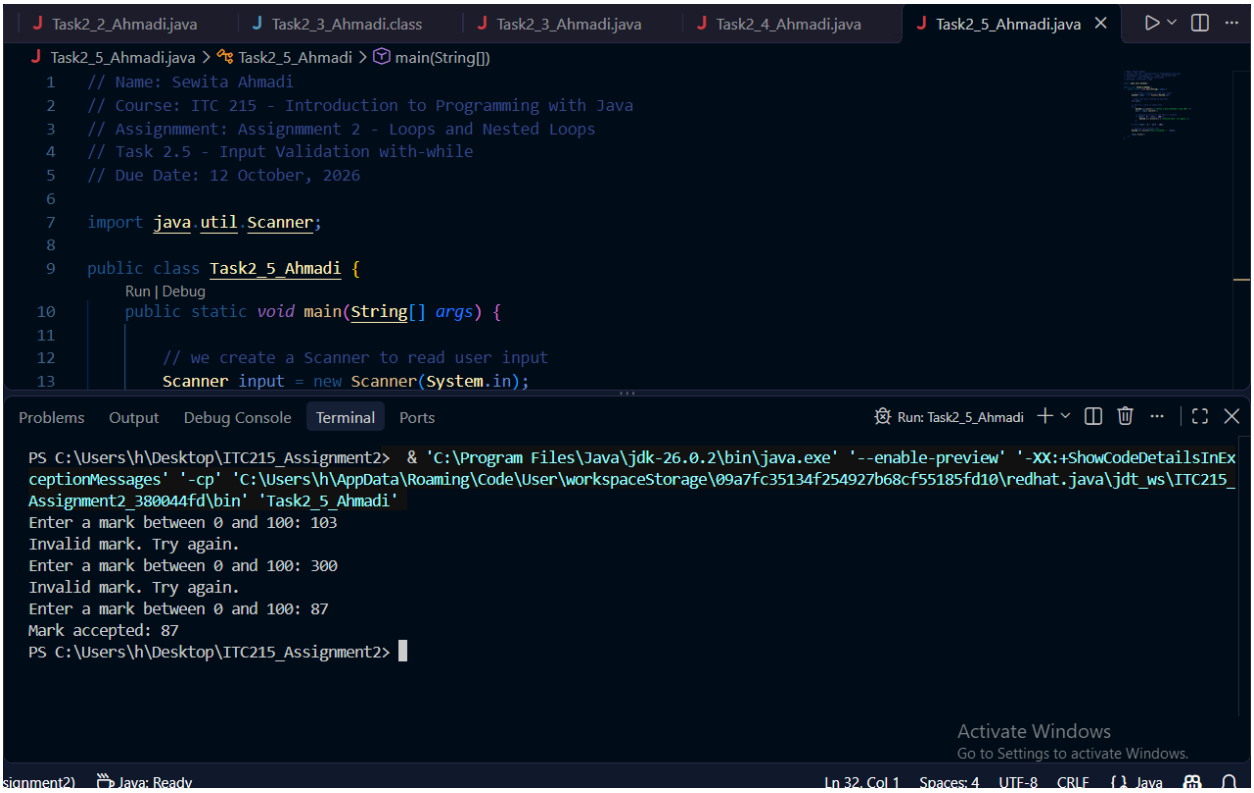


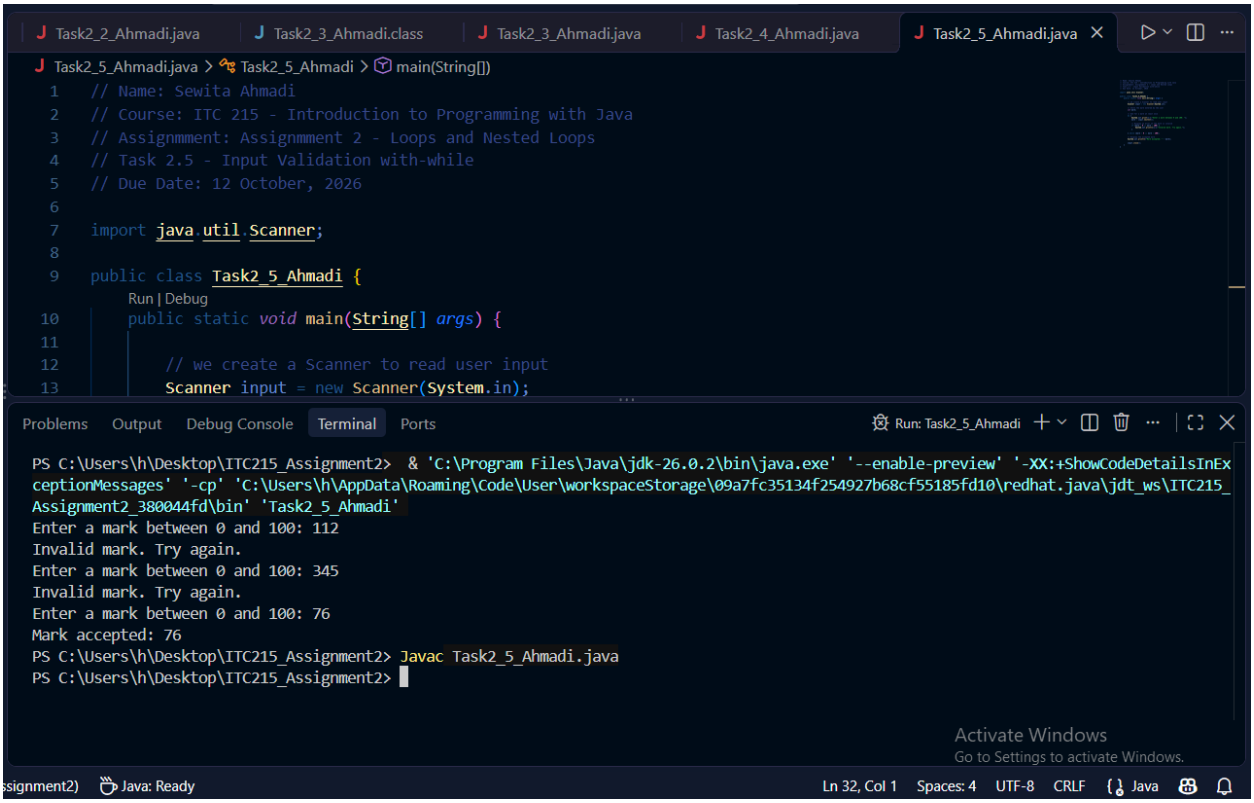
Task 2.4 Output screenshot





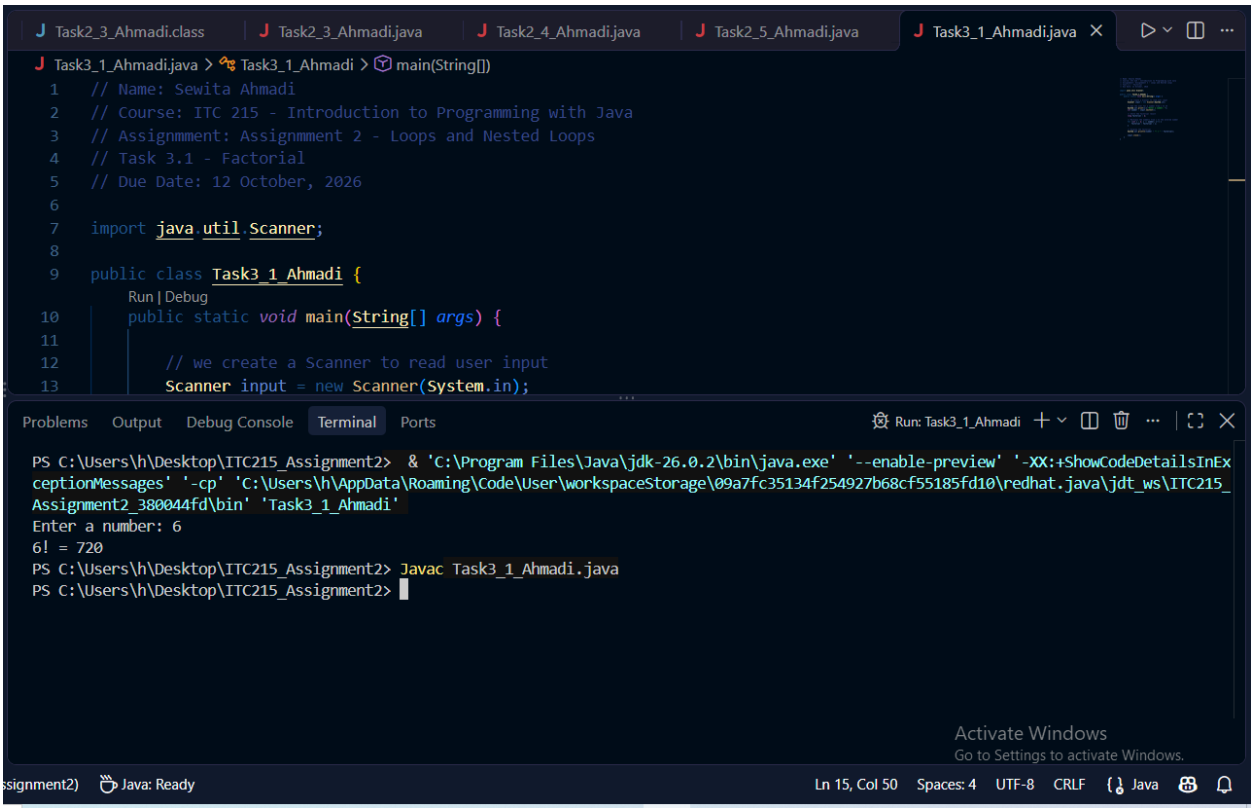
Task 2.5 Output Screenshots

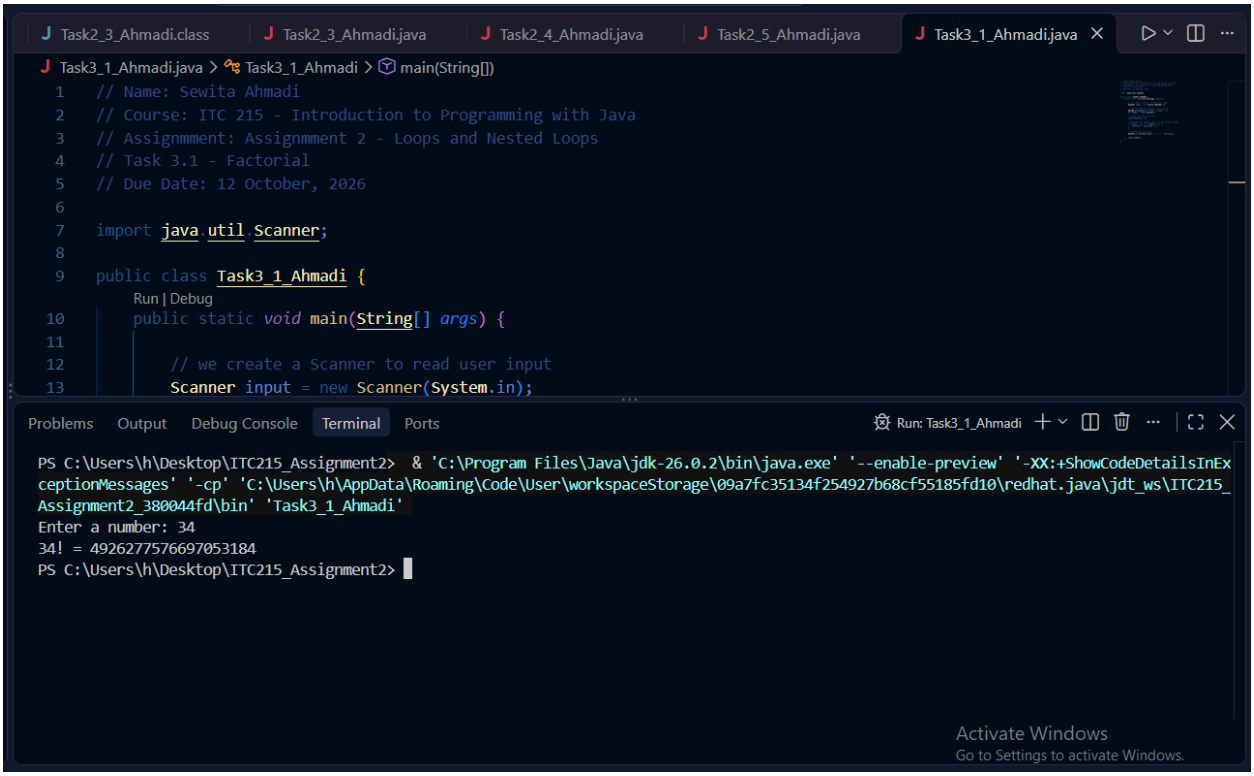

Task2_2_Ahmadi.java
Task2_3_Ahmadi.class
Task2_3_Ahmadi.java
Task2_4_Ahmadi.java
Task2_5_Ahmadi.java
Task2_5_Ahmadi.java > Task2_5_Ahmadi > main(String[])
// Name: Sewita Ahmadi
// Course: ITC 215 - Introduction to Programming with Java
// Assignmment: Assignmment 2 - Loops and Nested Loops
// Task 2.5 - Input Validation with-while
// Due Date: 12 October, 2026
import java.util.Scanner;
public class Task2_5_Ahmadi {
Run | Debug
public static void main(String[] args) {
// we create a Scanner to read user input
Scanner input = new Scanner(System.in);
Problems
Output
Debug Console
Terminal
Ports
Run: Task2_5_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task2_5_Ahmadi'
Enter a mark between 0 and 100: 103
Invalid mark. Try again.
Enter a mark between 0 and 100: 300
Invalid mark. Try again.
Enter a mark between 0 and 100: 87
Mark accepted: 87
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Java: Ready
Ln 32, Col 1
Spaces: 4
UTF-8
CRLF
Java


Task2_2_Ahmadi.java
Task2_3_Ahmadi.class
Task2_3_Ahmadi.java
Task2_4_Ahmadi.java
Task2_5_Ahmadi.java
Task2_5_Ahmadi.java > Task2_5_Ahmadi > main(String[])
// Name: Sewita Ahmadi
// Course: ITC 215 - Introduction to Programming with Java
// Assignmment: Assignmment 2 - Loops and Nested Loops
// Task 2.5 - Input Validation with-while
// Due Date: 12 October, 2026
import java.util.Scanner;
public class Task2_5_Ahmadi {
Run | Debug
public static void main(String[] args) {
// we create a Scanner to read user input
Scanner input = new Scanner(System.in);
Problems
Output
Debug Console
Terminal
Ports
Run: Task2_5_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task2_5_Ahmadi'
Enter a mark between 0 and 100: 112
Invalid mark. Try again.
Enter a mark between 0 and 100: 345
Invalid mark. Try again.
Enter a mark between 0 and 100: 76
Mark accepted: 76
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task2_5_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Java: Ready
Ln 32, Col 1
Spaces: 4
UTF-8
CRLF
Java

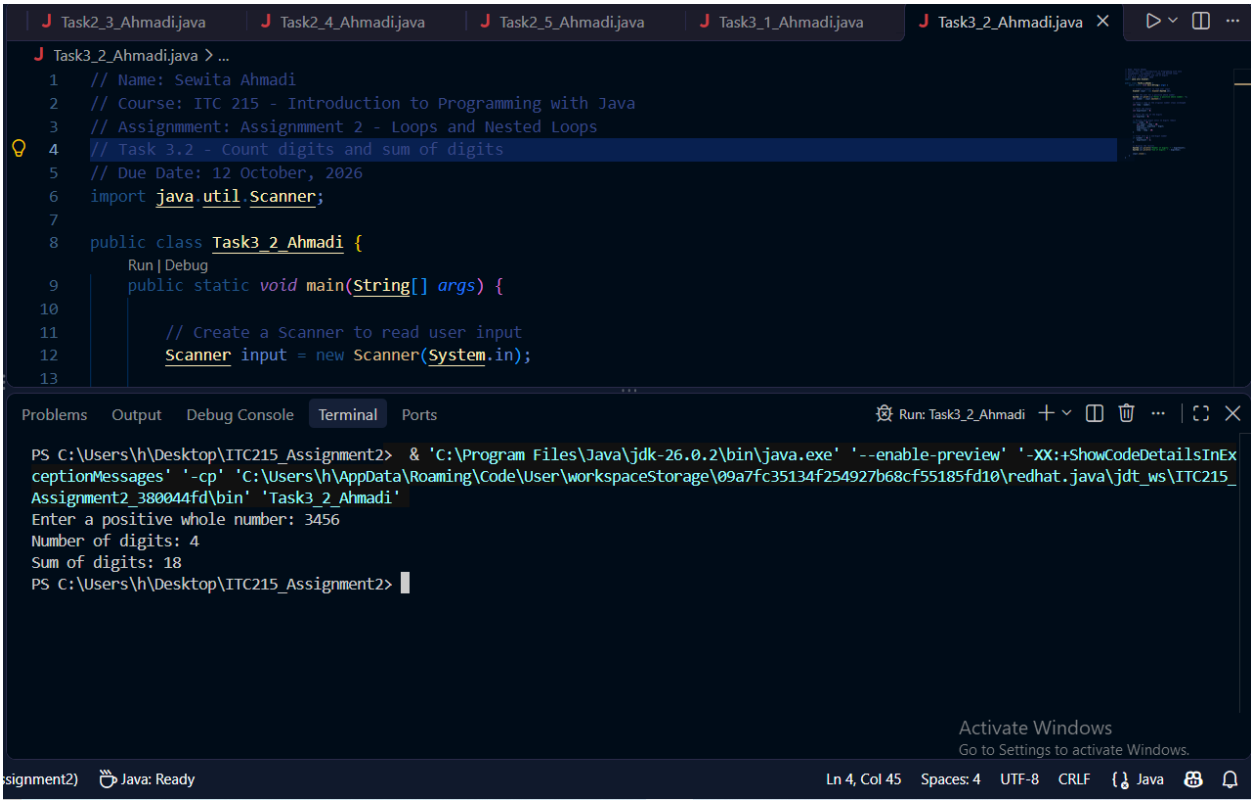
Part 3: Intermediate Loop Programs

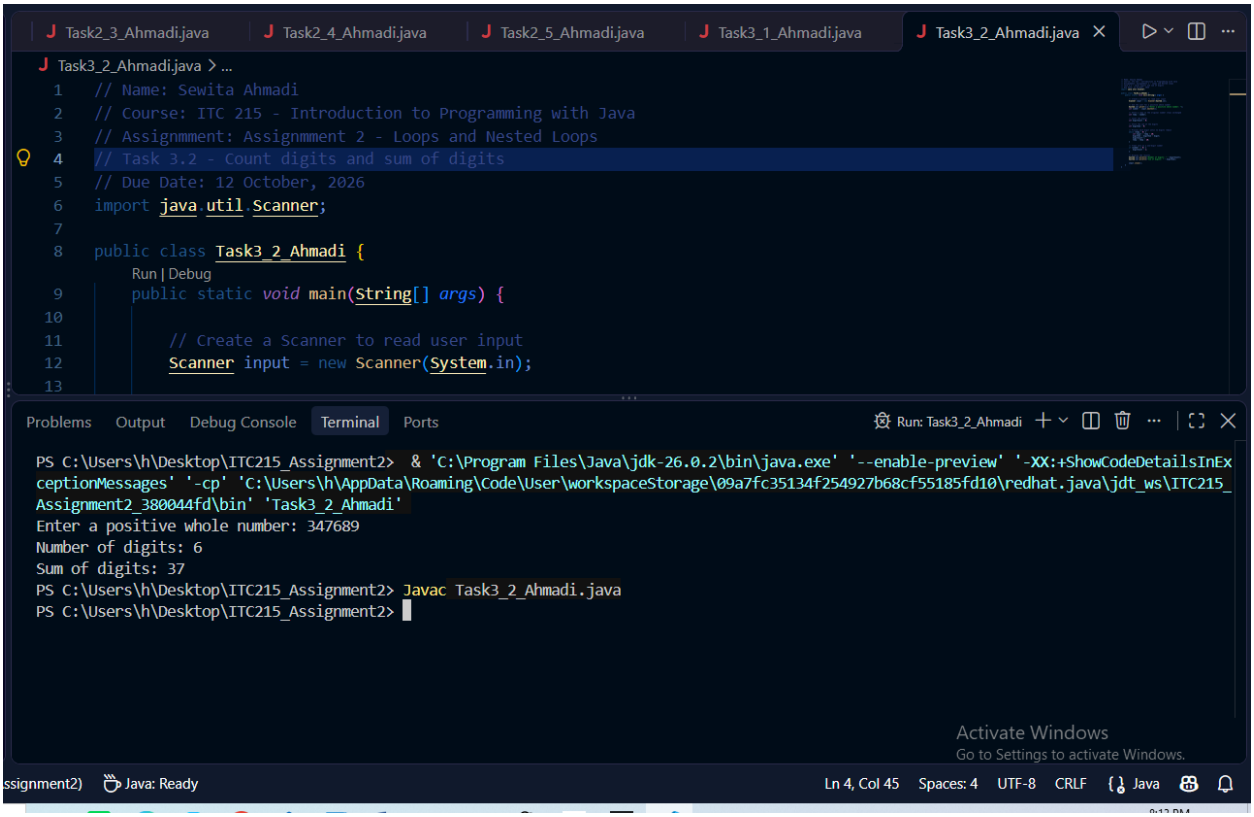
Task 3.1 Output Screenshot



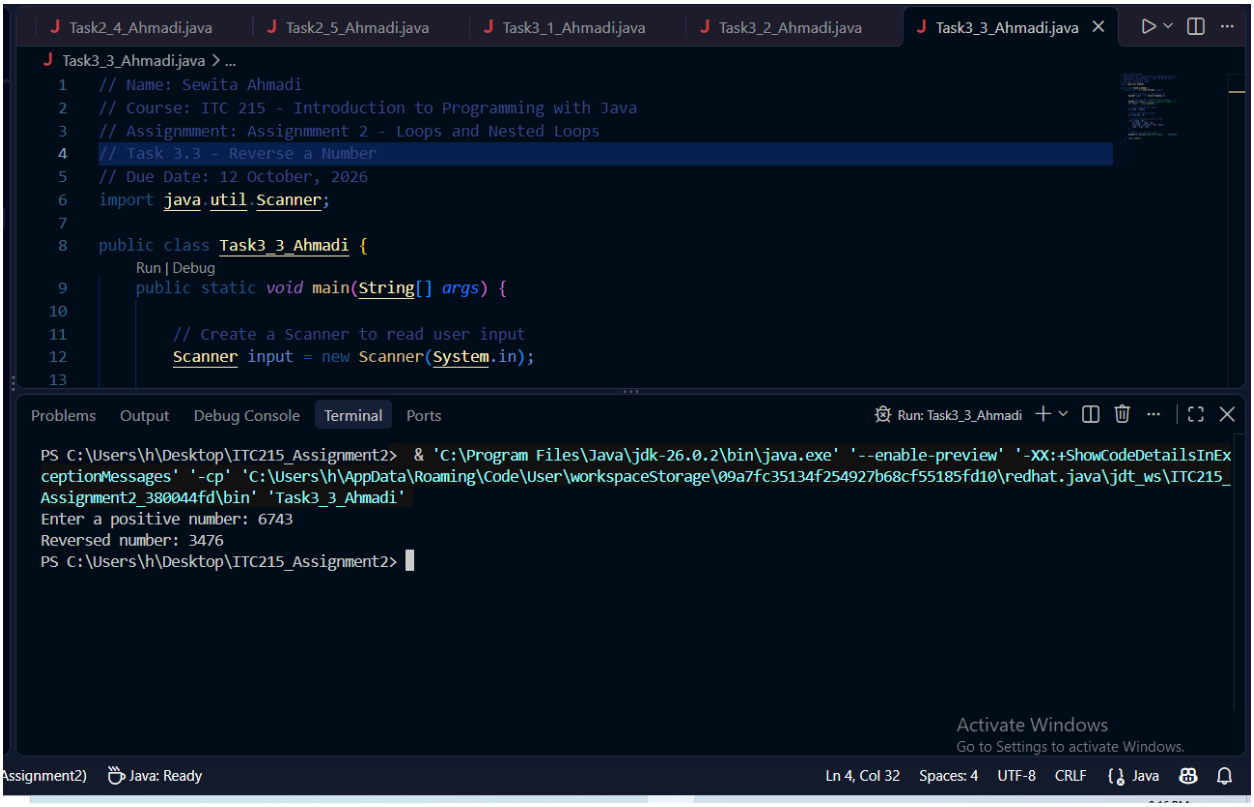


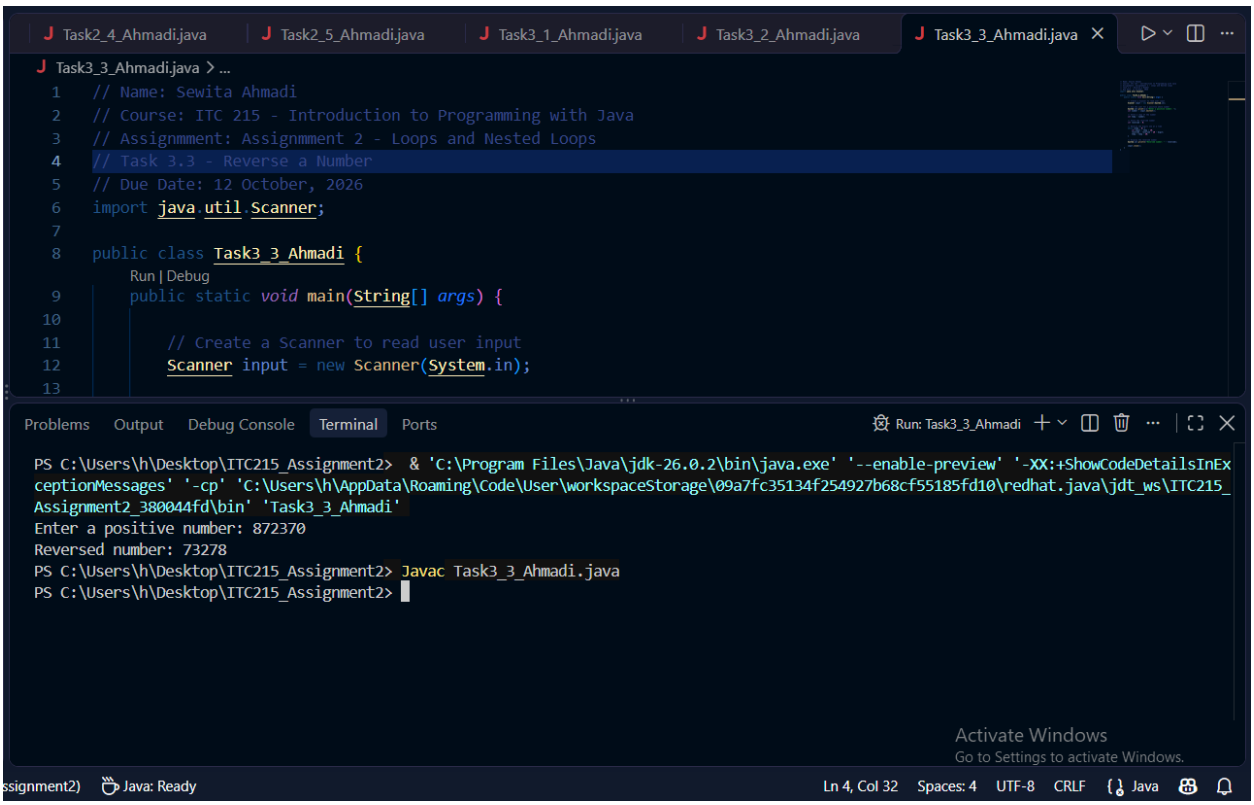
Task 3.2 Output Screenshots



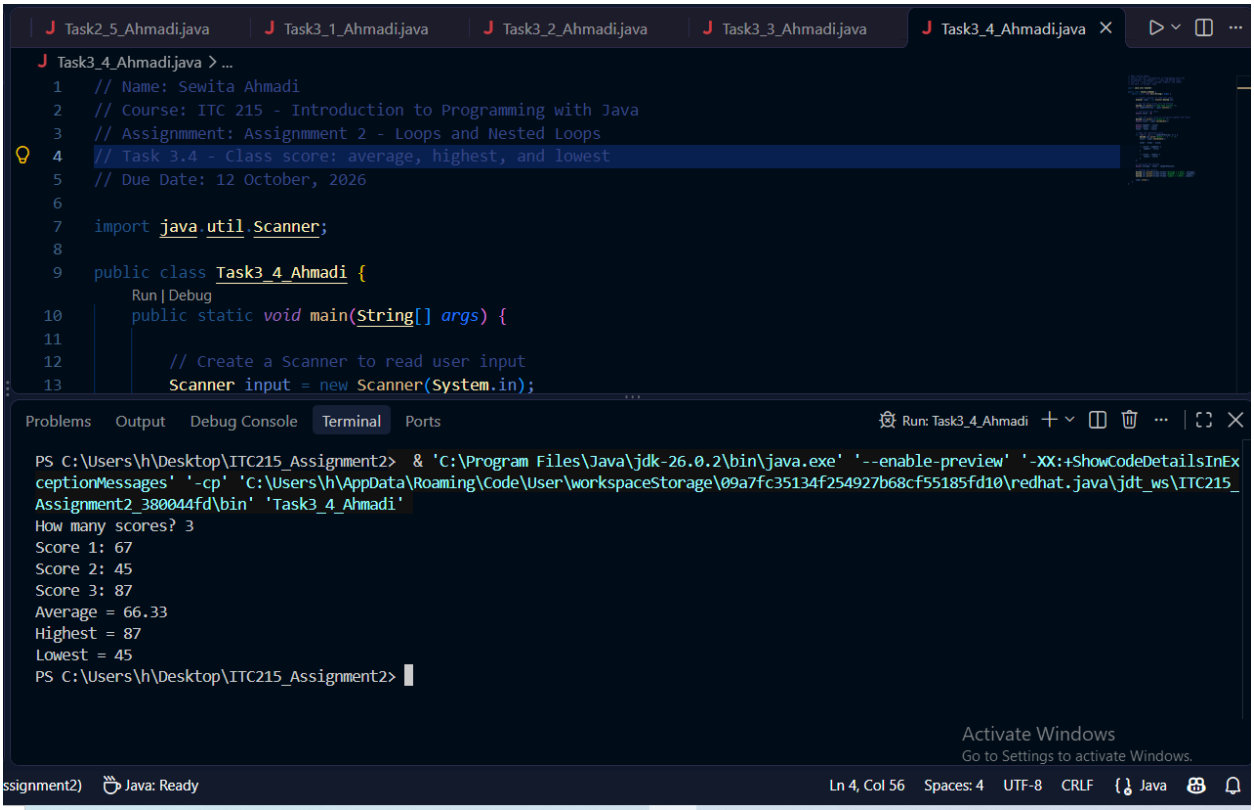


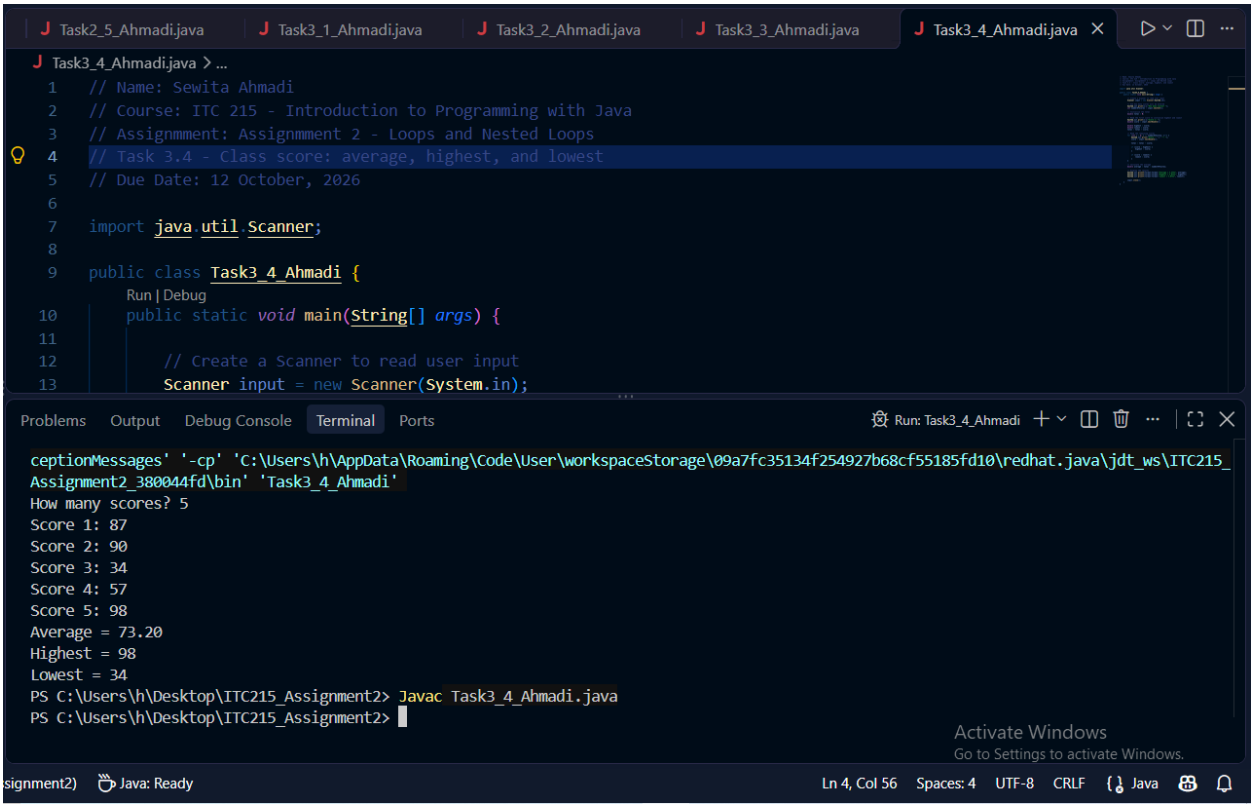
Task 3.3 Output screenshot


Task2_4_Ahmadi.java
Task2_5_Ahmadi.java
Task3_1_Ahmadi.java
Task3_2_Ahmadi.java
Task3_3_Ahmadi.java
Task3_3_Ahmadi.java > ...
1 // Name: Sewita Ahmadi
2 // Course: ITC 215 - Introduction to Programming with Java
3 // Assignmment: Assignmment 2 - Loops and Nested Loops
4 // Task 3.3 - Reverse a Number
5 // Due Date: 12 October, 2026
6 import java.util.Scanner;
7
8 public class Task3_3_Ahmadi {
Run | Debug
9 public static void main(String[] args) {
10
11 // Create a Scanner to read user input
12 Scanner input = new Scanner(System.in);
13
Problems
Output
Debug Console
Terminal
Ports
Run: Task3_3_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task3_3_Ahmadi'
Enter a positive number: 6743
Reversed number: 3476
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Assignment2)
Java: Ready
Ln 4, Col 32
Spaces: 4
UTF-8
CRLF
Java


Task2_4_Ahmadi.java
Task2_5_Ahmadi.java
Task3_1_Ahmadi.java
Task3_2_Ahmadi.java
Task3_3_Ahmadi.java
Task3_3_Ahmadi.java > ...
1 // Name: Sewita Ahmadi
2 // Course: ITC 215 - Introduction to Programming with Java
3 // Assignmment: Assignmment 2 - Loops and Nested Loops
4 // Task 3.3 - Reverse a Number
5 // Due Date: 12 October, 2026
6 import java.util.Scanner;
7
8 public class Task3_3_Ahmadi {
Run | Debug
9 public static void main(String[] args) {
10
11 // Create a Scanner to read user input
12 Scanner input = new Scanner(System.in);
13
Problems
Output
Debug Console
Terminal
Ports
Run: Task3_3_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task3_3_Ahmadi'
Enter a positive number: 872370
Reversed number: 73278
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task3_3_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
ssignment2)
Java: Ready
Ln 4, Col 32
Spaces: 4
UTF-8
CRLF
Java

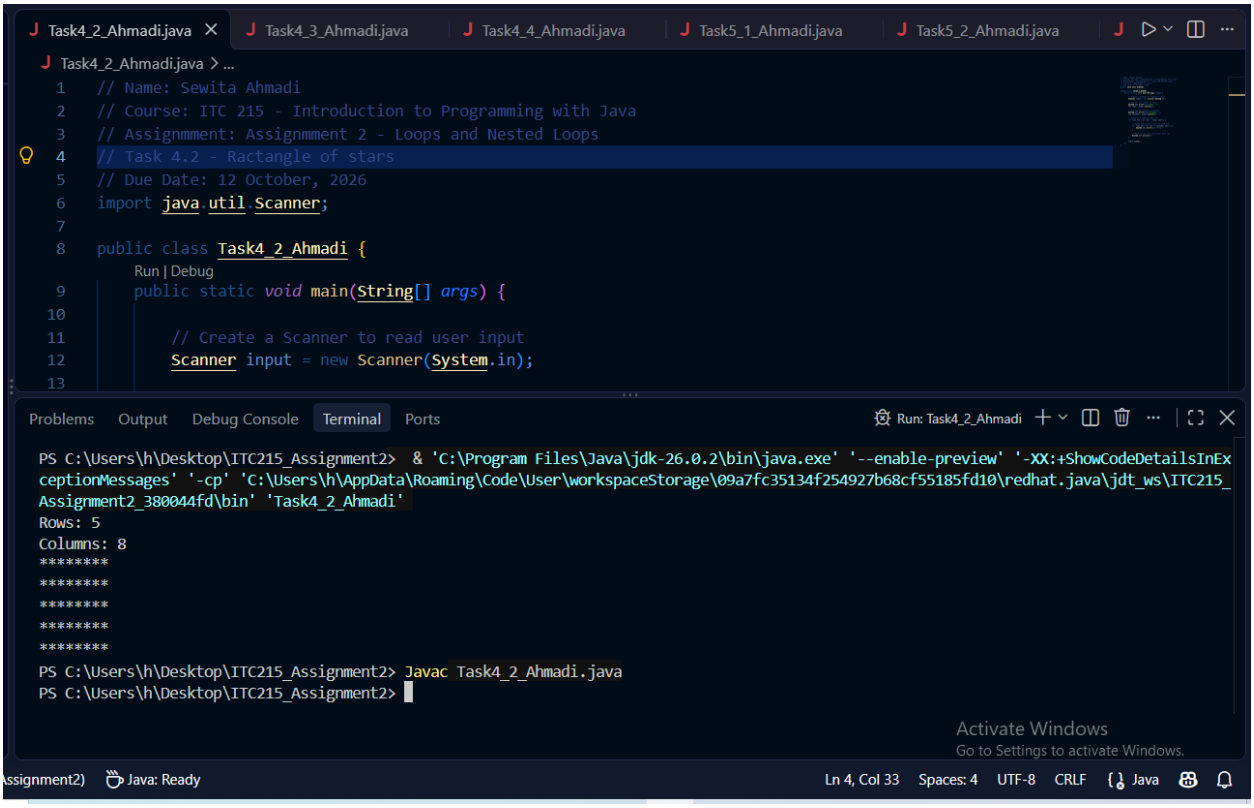
Task 3.4 Output Screenshot

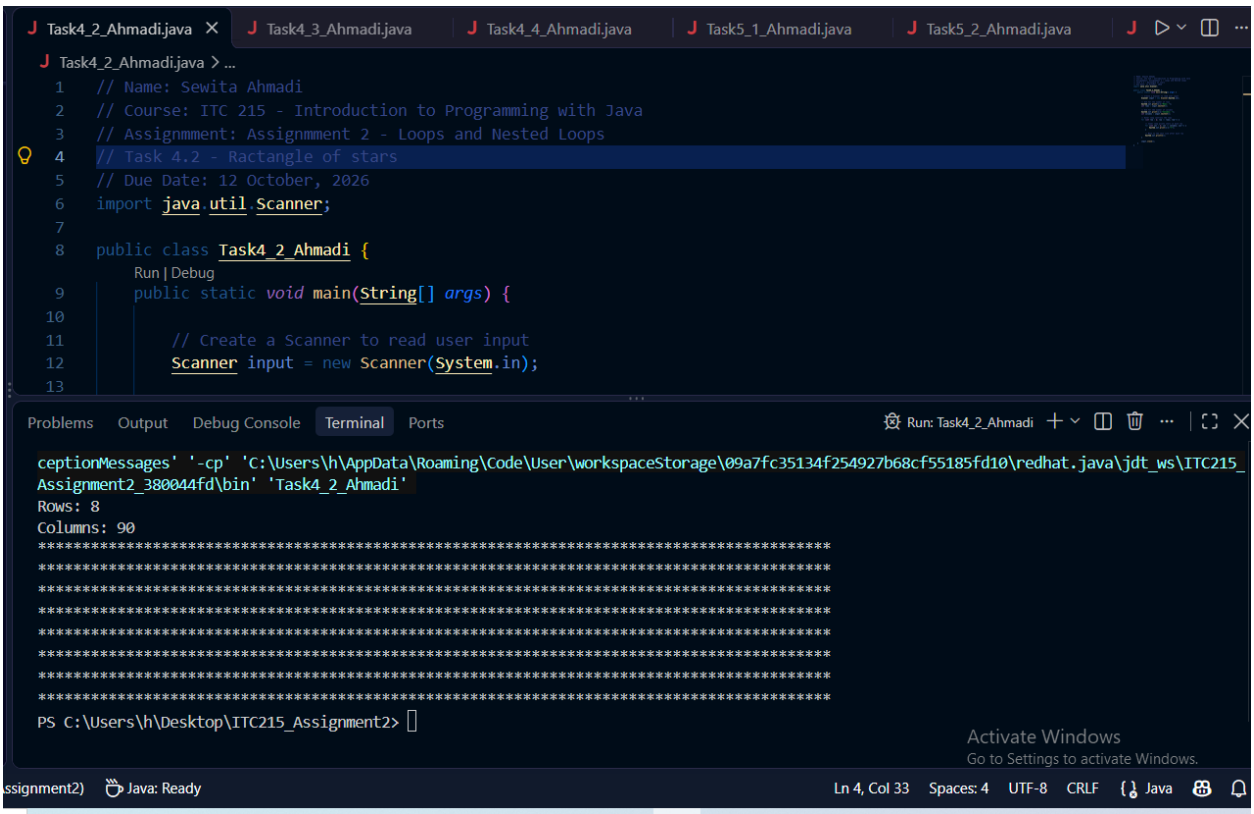




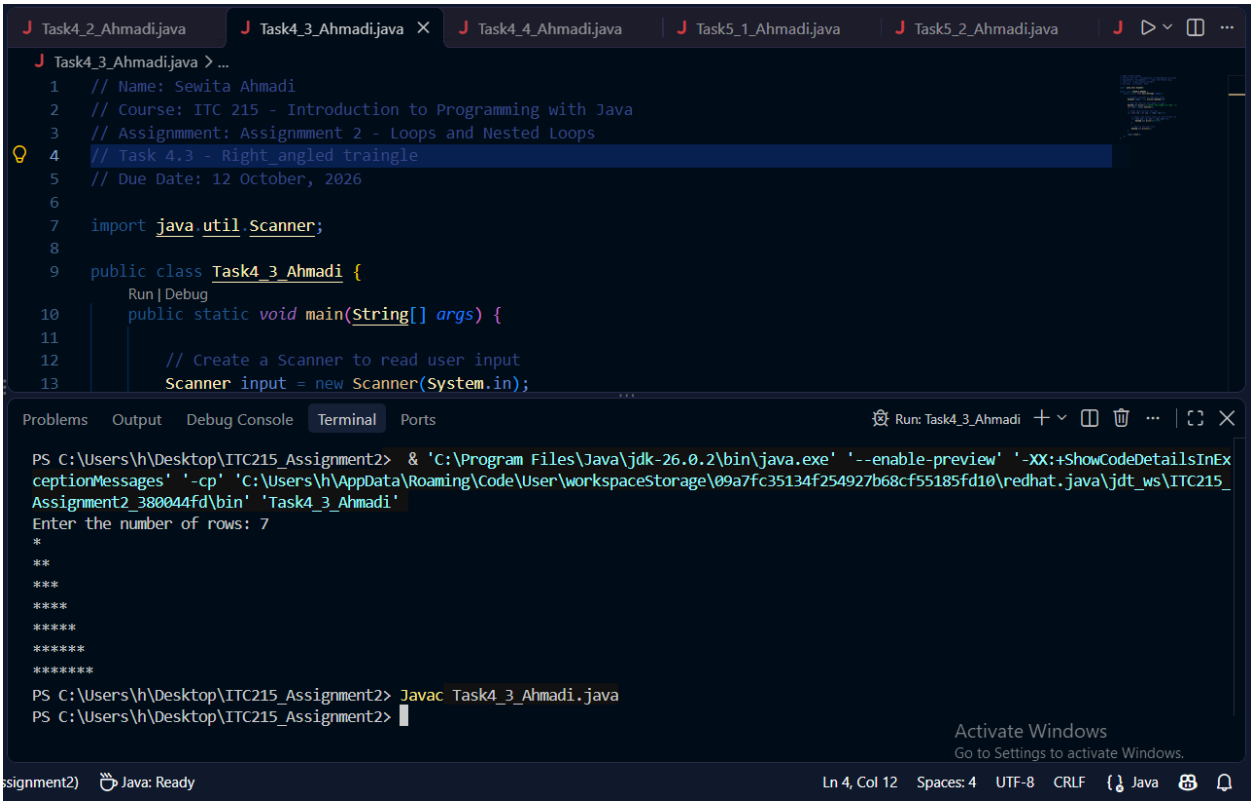
Part 4: Introduction to Nested Loops

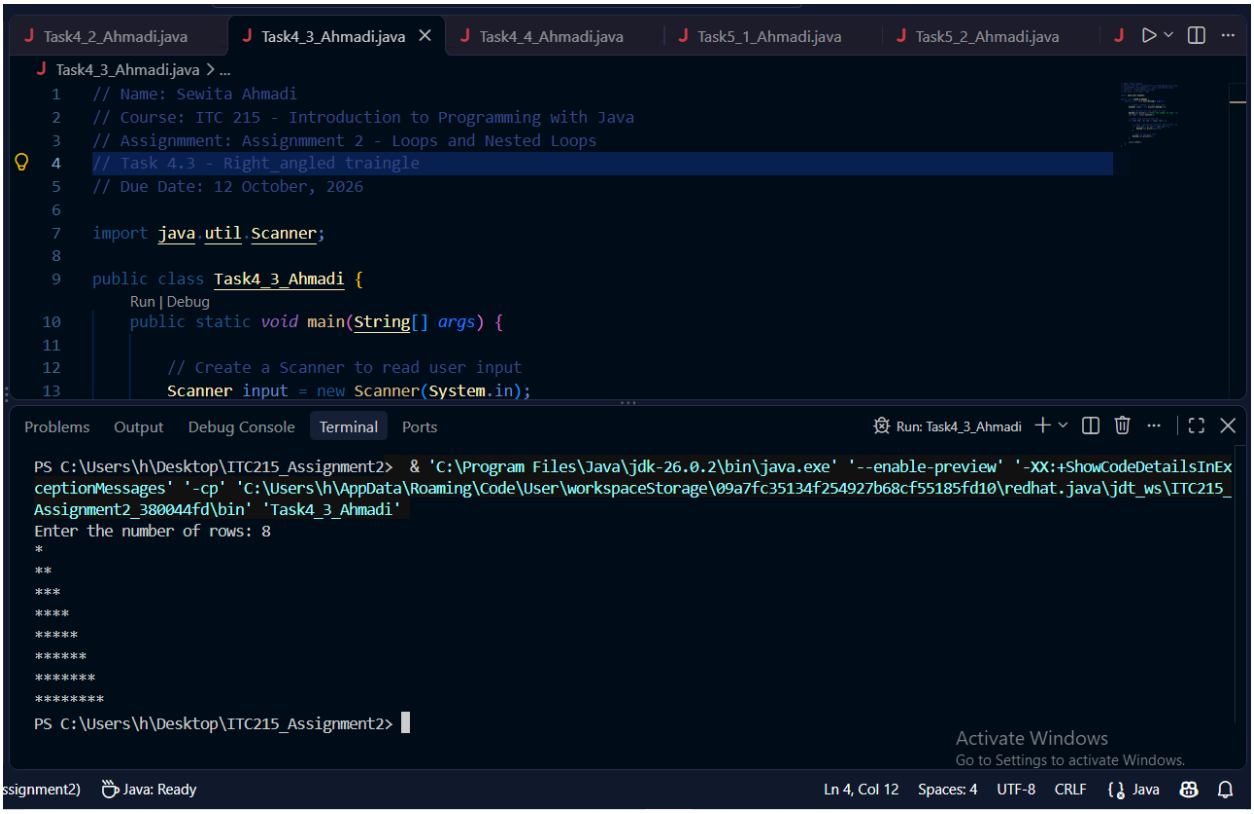
Task 4.2 Output Screenshot





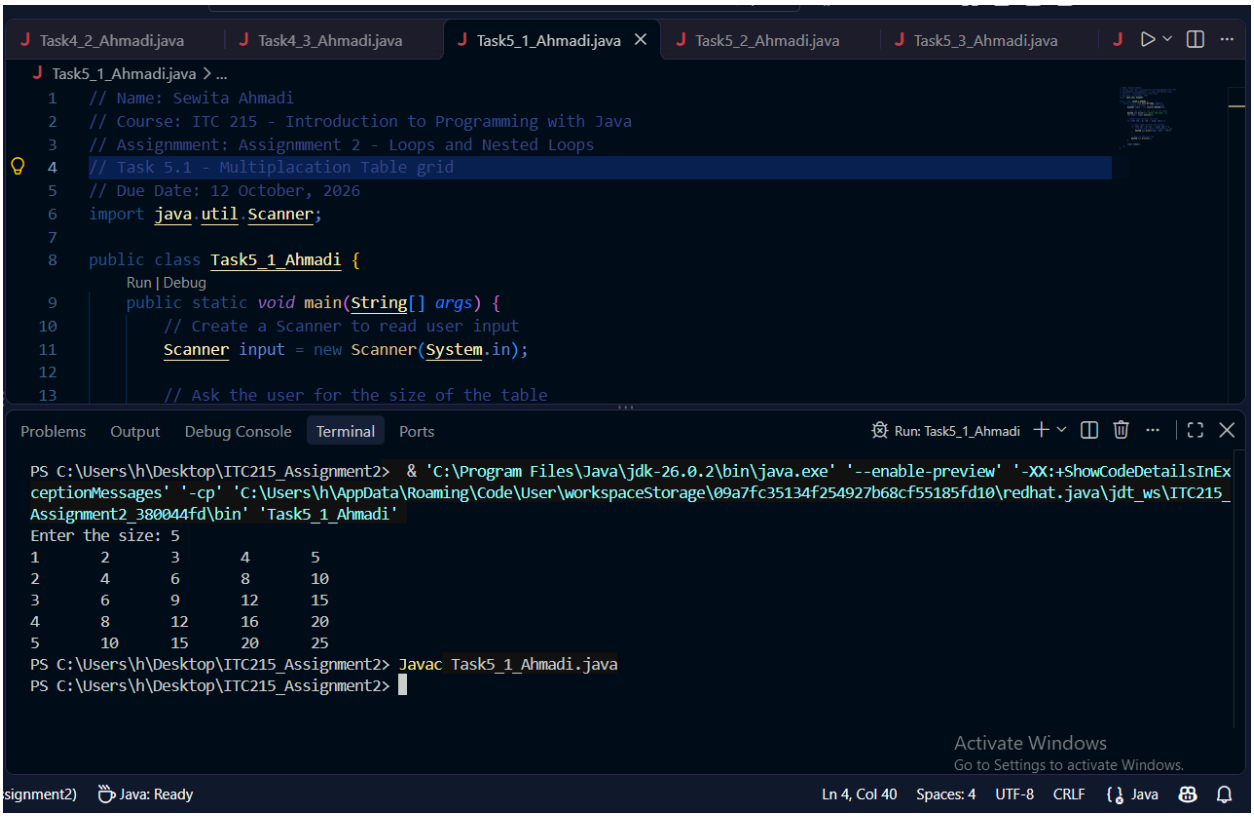
Task 4.3 Output Screenshot

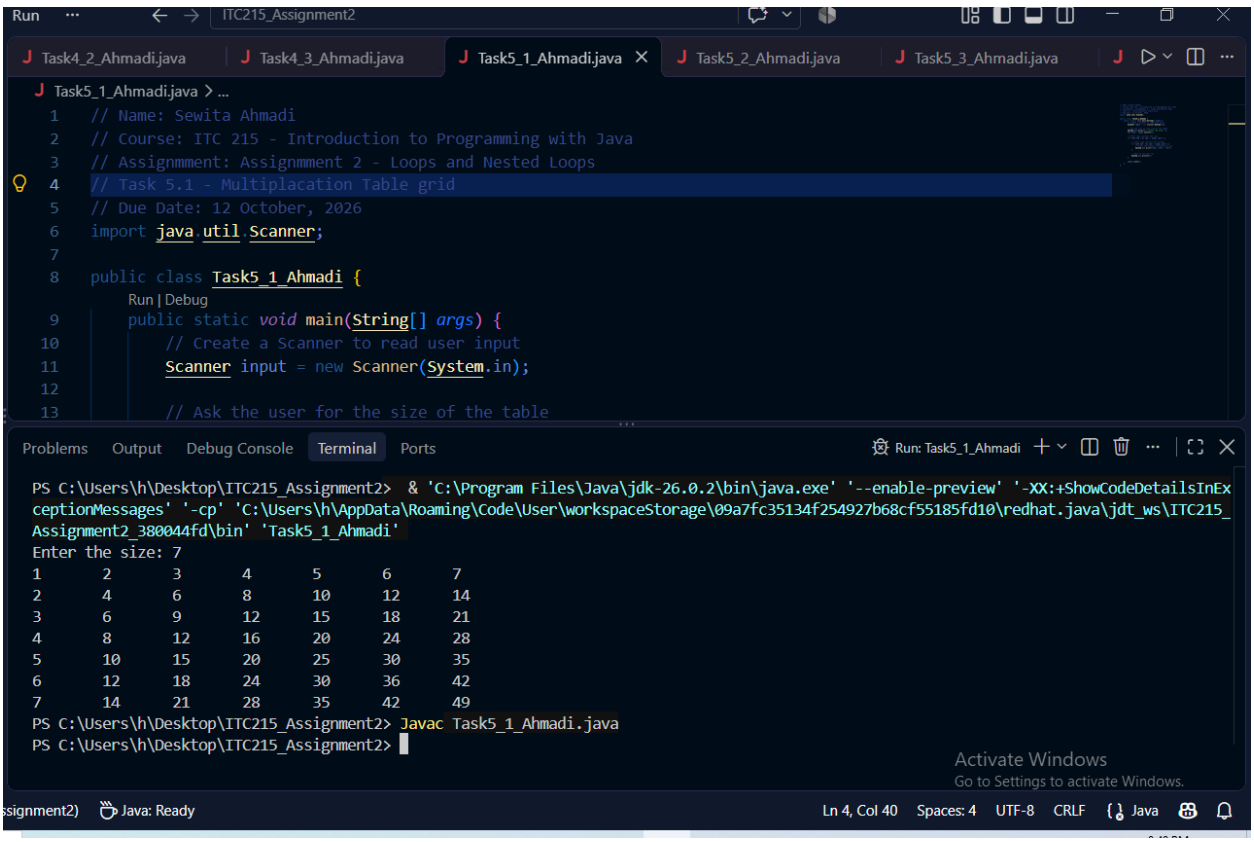




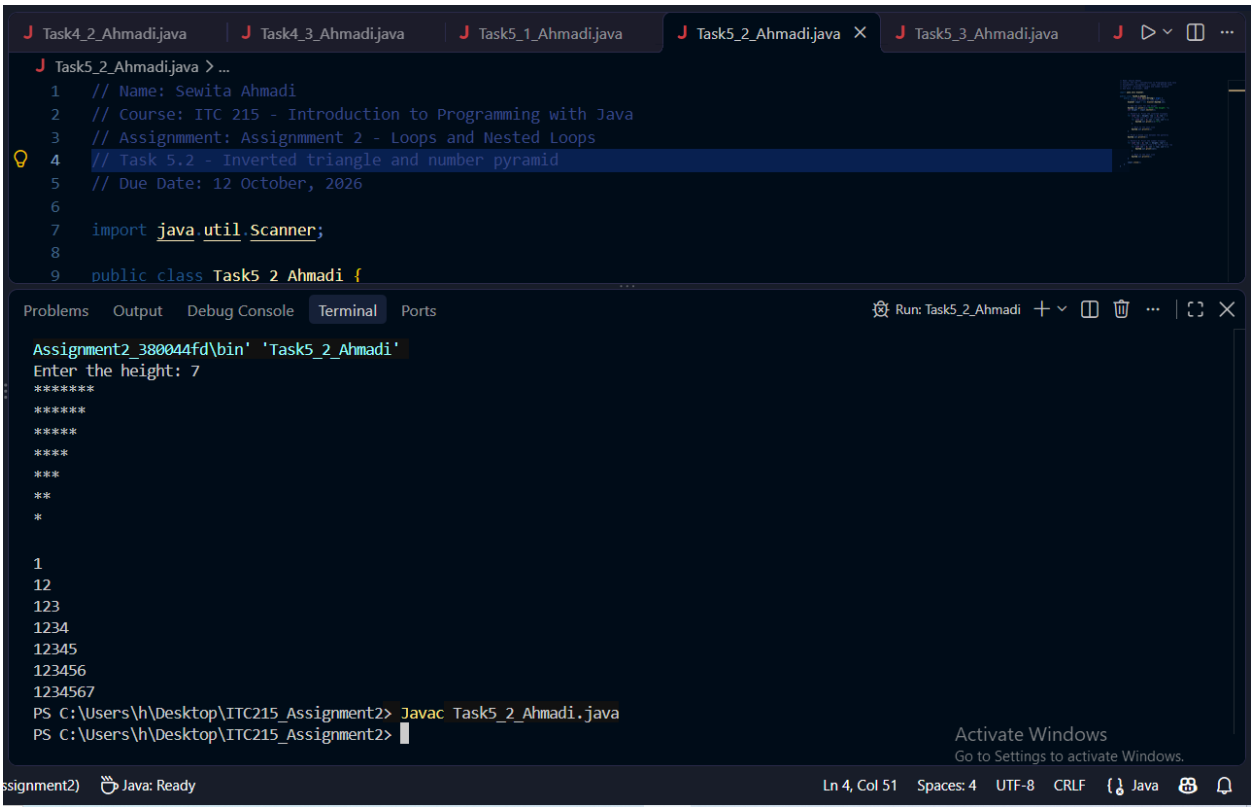
Part 5 - Advanced Nested Loops: Tables and patterns

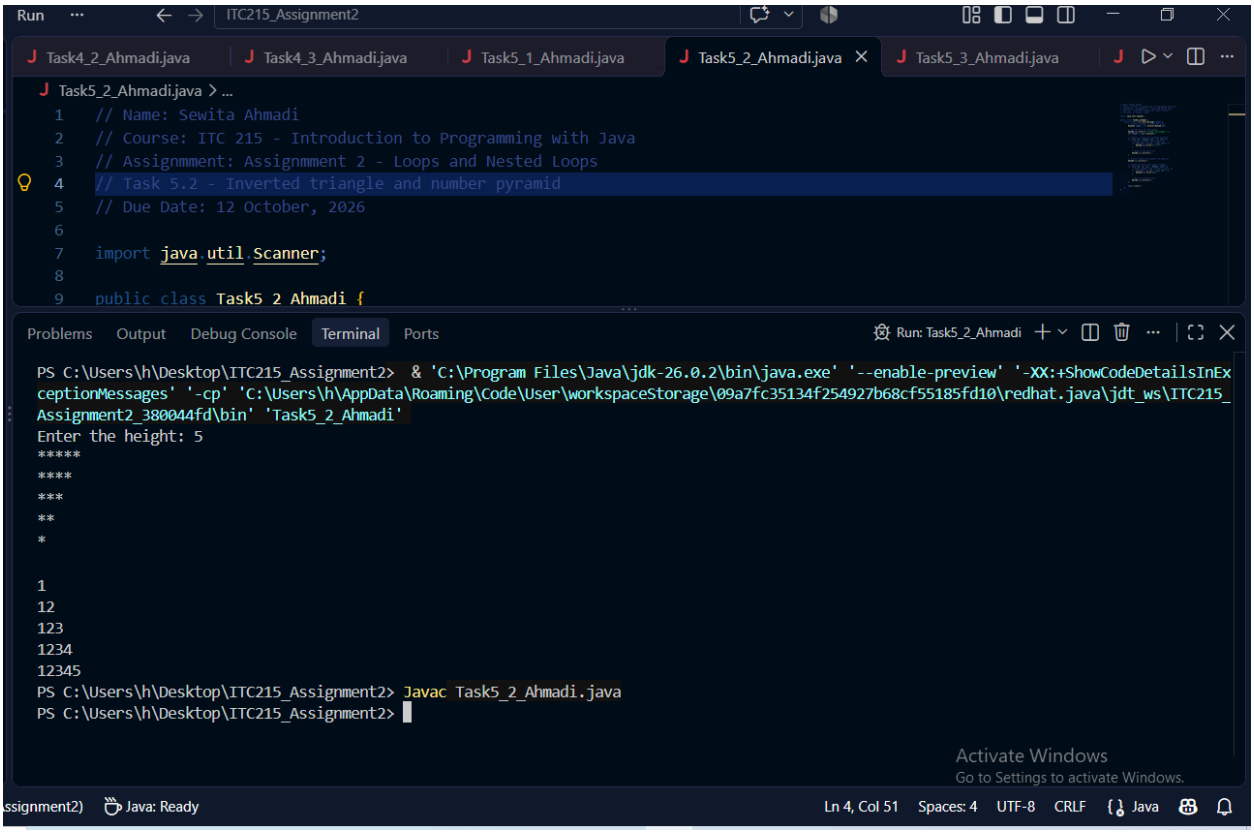
Task 5.1 Output Screenshot


Task4_2_Ahmadi.java
Task4_3_Ahmadi.java
Task5_1_Ahmadi.java
Task5_2_Ahmadi.java
Task5_3_Ahmadi.java
// Name: Sewita Ahmadi
// Course: ITC 215 - Introduction to Programming with Java
// Assignmment: Assignmment 2 - Loops and Nested Loops
// Task 5.1 - Multiplacation Table grid
// Due Date: 12 October, 2026
import java.util.Scanner;
public class Task5_1_Ahmadi {
Run | Debug
public static void main(String[] args) {
// Create a Scanner to read user input
Scanner input = new Scanner(System.in);
// Ask the user for the size of the table
Problems
Output
Debug Console
Terminal
Ports
Run: Task5_1_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task5_1_Ahmadi'
Enter the size: 5
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
4 8 12 16 20
5 10 15 20 25
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task5_1_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Java: Ready
Ln 4, Col 40
Spaces: 4
UTF-8
CRLF
Java

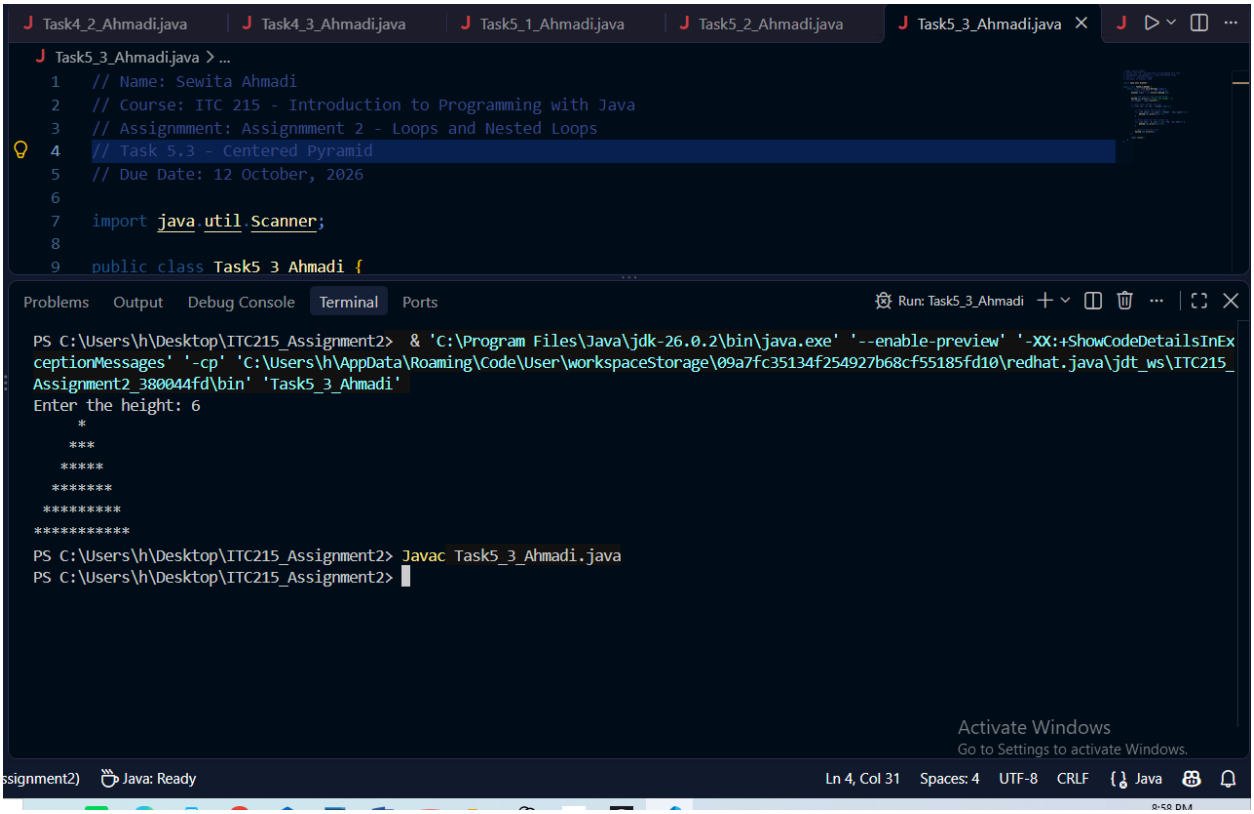

Run
ITC215_Assignment2
Task4_2_Ahmadi.java
Task4_3_Ahmadi.java
Task5_1_Ahmadi.java
Task5_2_Ahmadi.java
Task5_3_Ahmadi.java
// Name: Sewita Ahmadi
// Course: ITC 215 - Introduction to Programming with Java
// Assignmment: Assignmment 2 - Loops and Nested Loops
// Task 5.1 - Multiplacation Table grid
// Due Date: 12 October, 2026
import java.util.Scanner;
public class Task5_1_Ahmadi {
Run | Debug
public static void main(String[] args) {
// Create a Scanner to read user input
Scanner input = new Scanner(System.in);
// Ask the user for the size of the table
Problems
Output
Debug Console
Terminal
Ports
Run: Task5_1_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task5_1_Ahmadi'
Enter the size: 7
1 2 3 4 5 6 7
2 4 6 8 10 12 14
3 6 9 12 15 18 21
4 8 12 16 20 24 28
5 10 15 20 25 30 35
6 12 18 24 30 36 42
7 14 21 28 35 42 49
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task5_1_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Java: Ready
Ln 4, Col 40
Spaces: 4
UTF-8
CRLF
Java

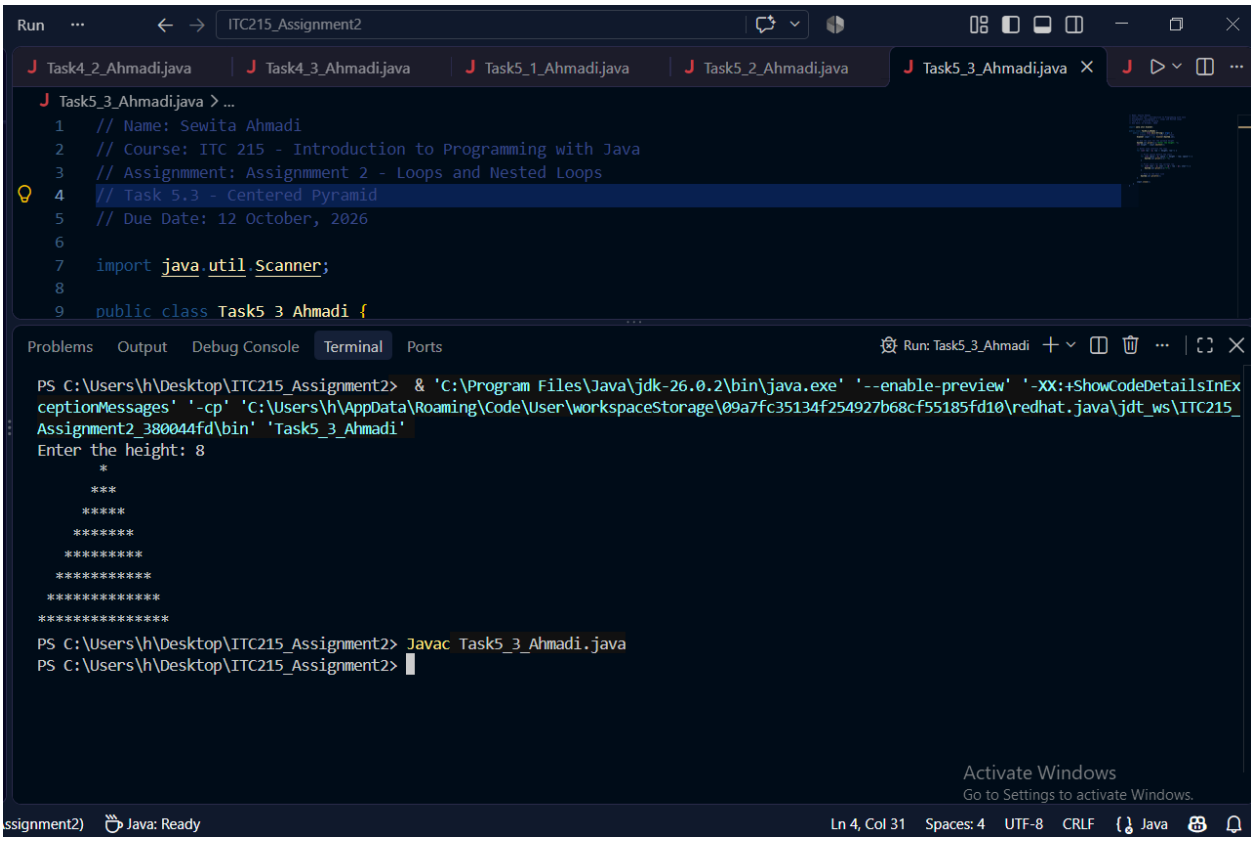
Task 5.2 Output



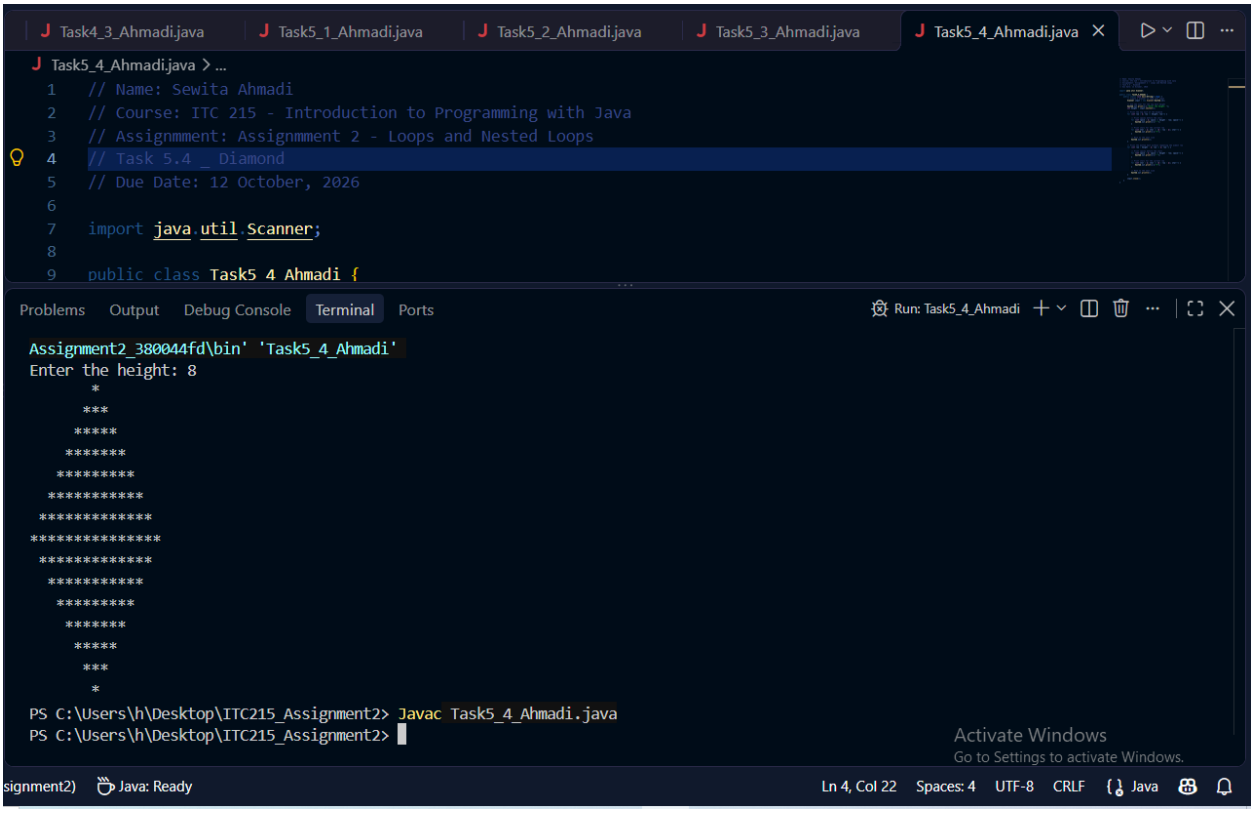


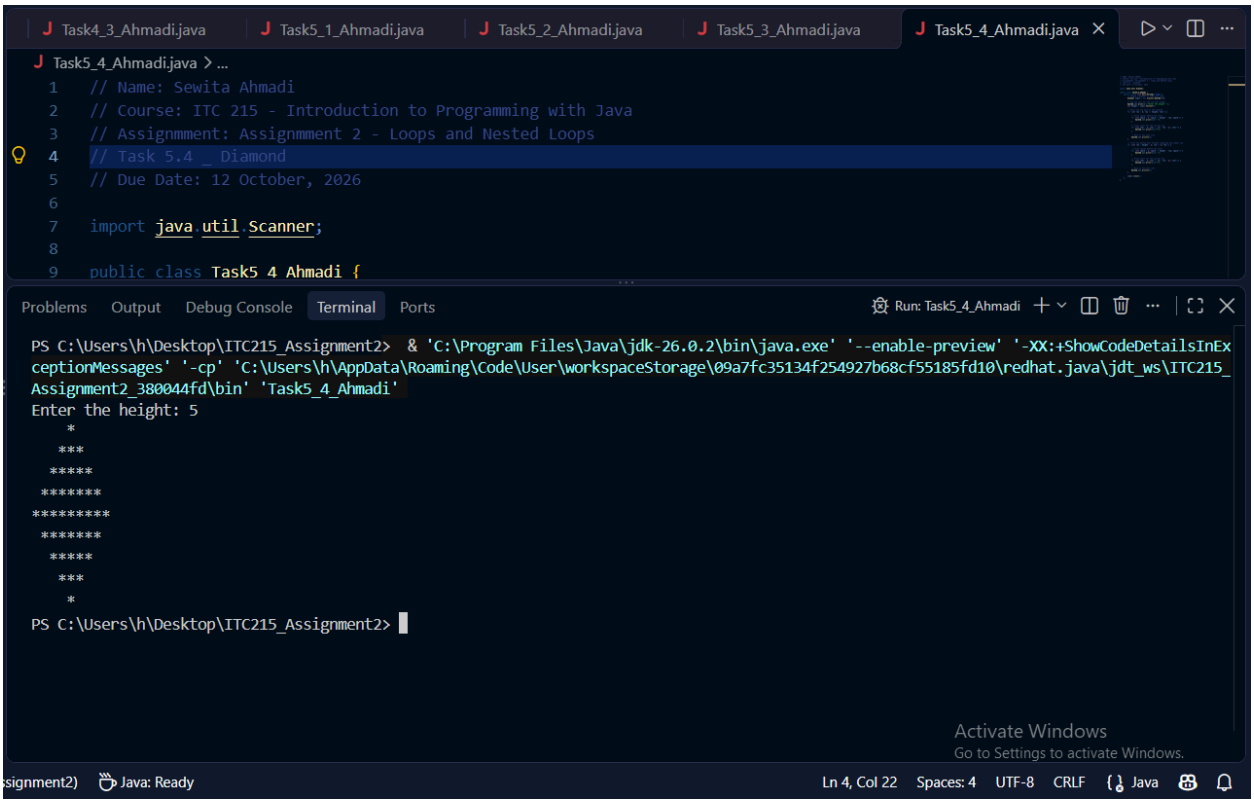
Task 5.3 Output screenshot


Task4_2_Ahmadi.java
Task4_3_Ahmadi.java
Task5_1_Ahmadi.java
Task5_2_Ahmadi.java
Task5_3_Ahmadi.java
// Name: Sewita Ahmadi
// Course: ITC 215 - Introduction to Programming with Java
// Assignmment: Assignmment 2 - Loops and Nested Loops
// Task 5.3 - Centered Pyramid
// Due Date: 12 October, 2026
import java.util.Scanner;
public class Task5_3_Ahmadi {
Problems
Output
Debug Console
Terminal
Ports
Run: Task5_3_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task5_3_Ahmadi'
Enter the height: 6
*
***
*****
****
****
****
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task5_3_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Java: Ready
Ln 4, Col 31
Spaces: 4
UTF-8
CRLF
Java


Run
ITC215_Assignment2
Task4_2_Ahmadi.java
Task4_3_Ahmadi.java
Task5_1_Ahmadi.java
Task5_2_Ahmadi.java
Task5_3_Ahmadi.java
// Name: Sewita Ahmadi
// Course: ITC 215 - Introduction to Programming with Java
// Assignmment: Assignmment 2 - Loops and Nested Loops
// Task 5.3 - Centered Pyramid
// Due Date: 12 October, 2026
import java.util.Scanner;
public class Task5_3_Ahmadi {
Problems
Output
Debug Console
Terminal
Ports
Run: Task5_3_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task5_3_Ahmadi'
Enter the height: 8
*
***
*****
****
****
****
****
****
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task5_3_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
Java: Ready
Ln 4, Col 31
Spaces: 4
UTF-8
CRLF
Java

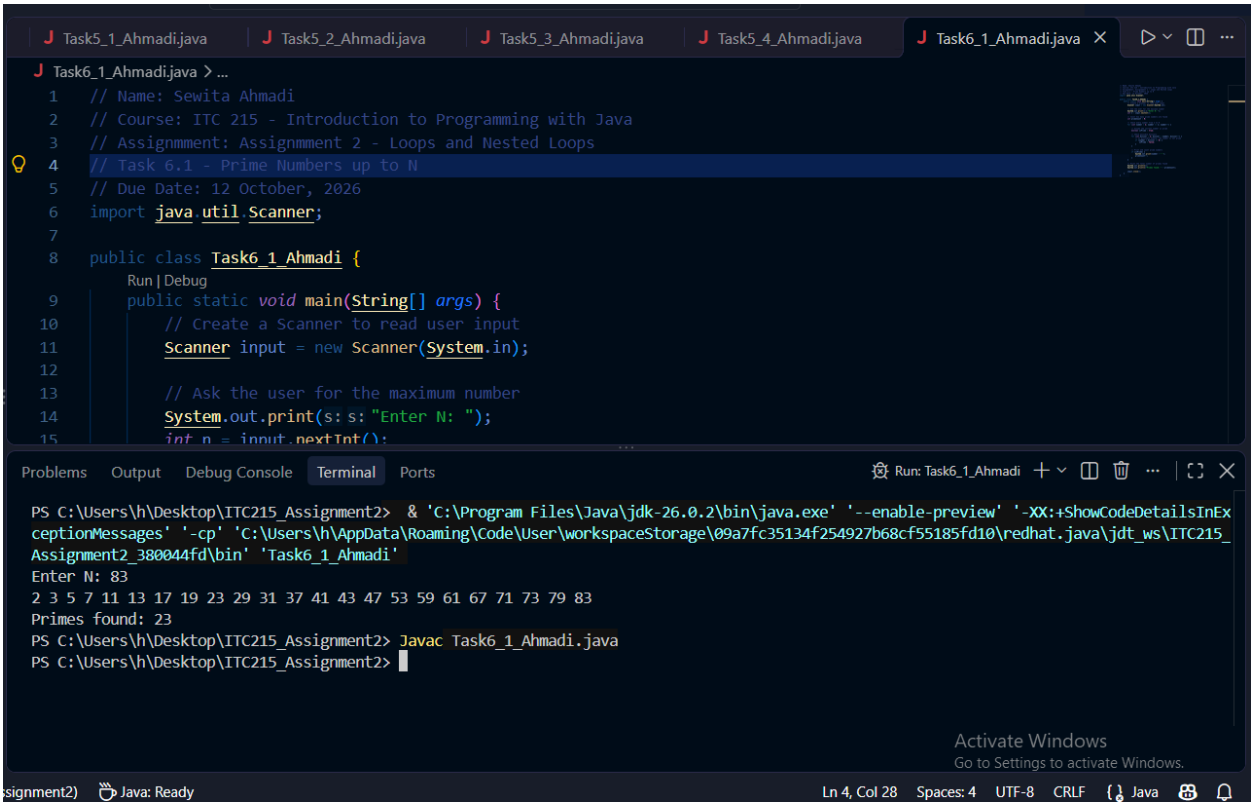
Task 5.4 Output screenshot

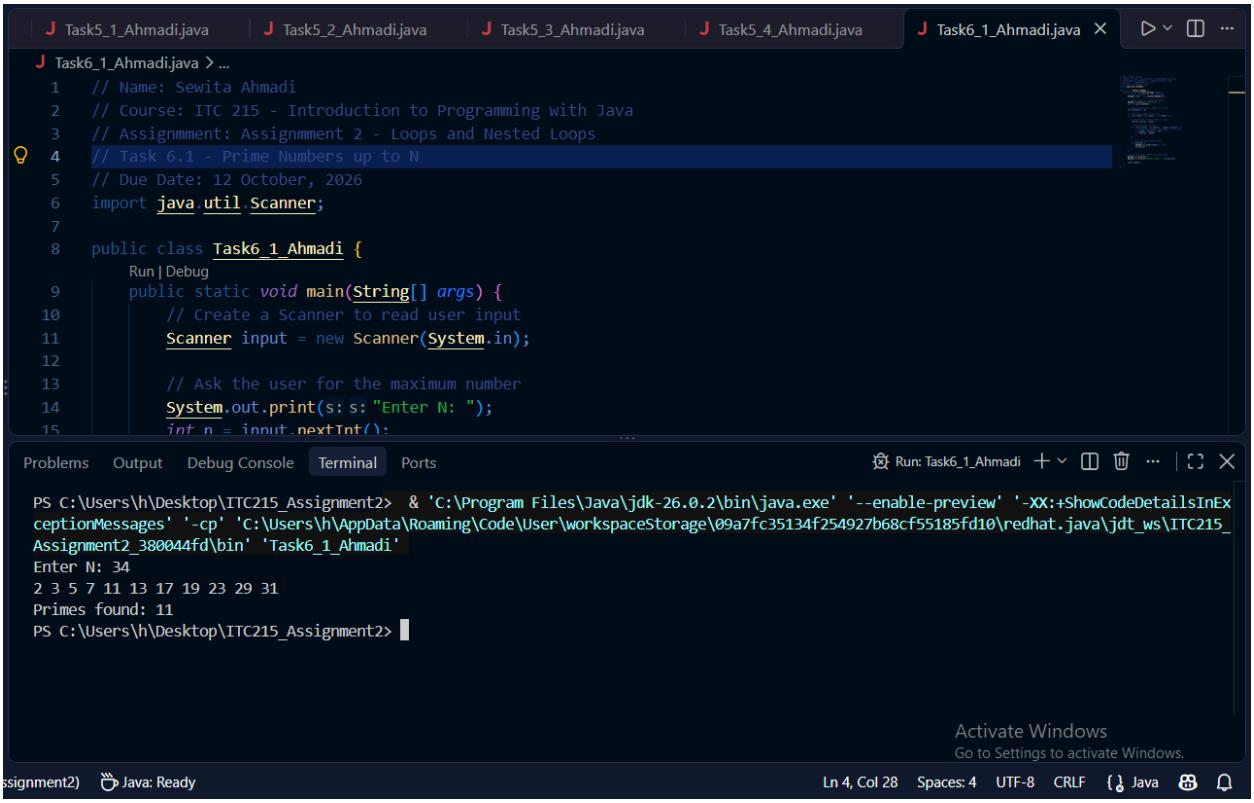




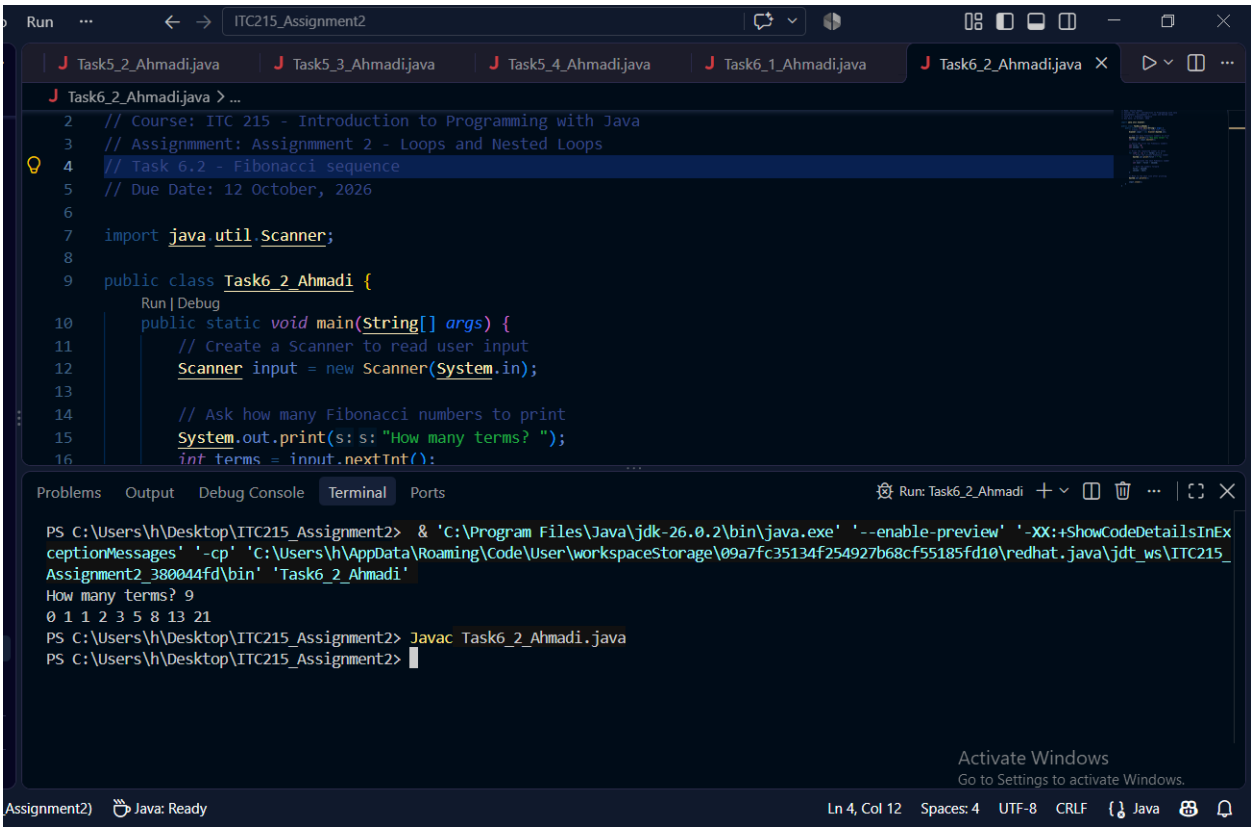
Part 6: Challanging Problems

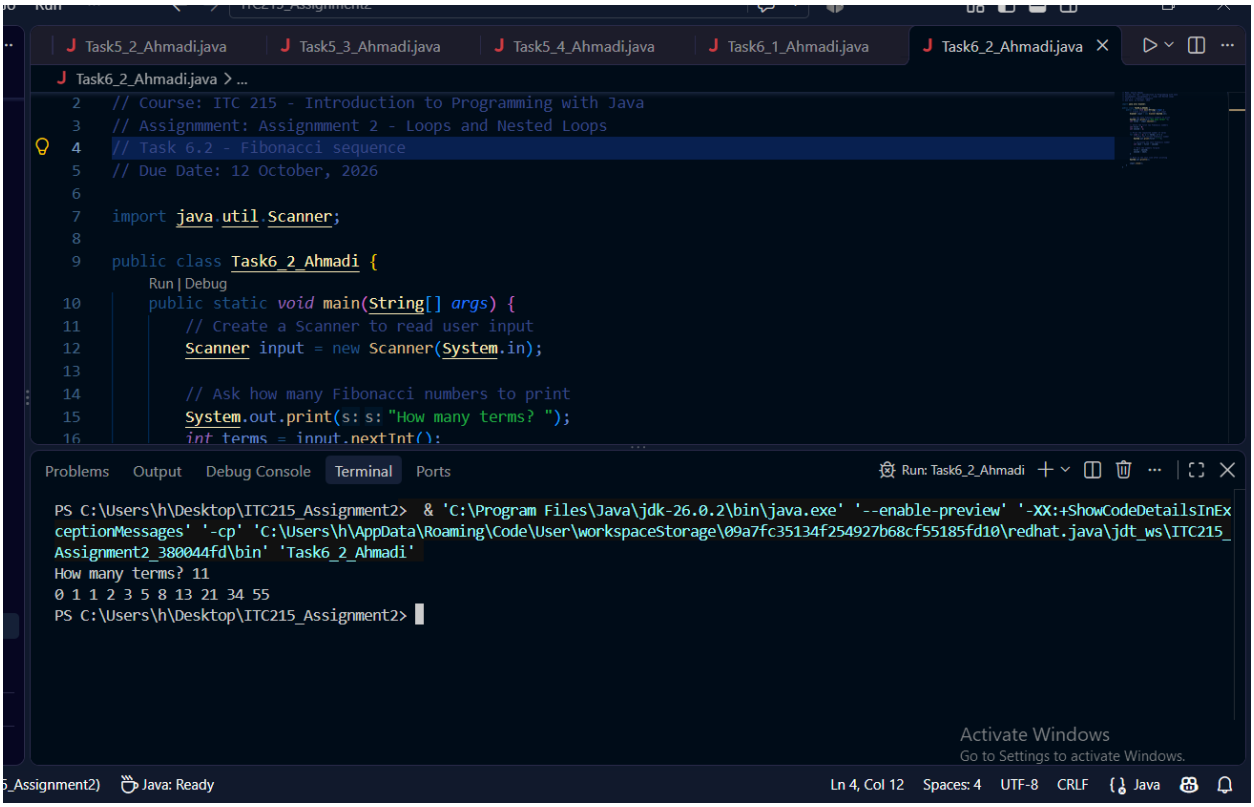
Task 6.1 Output screenshots



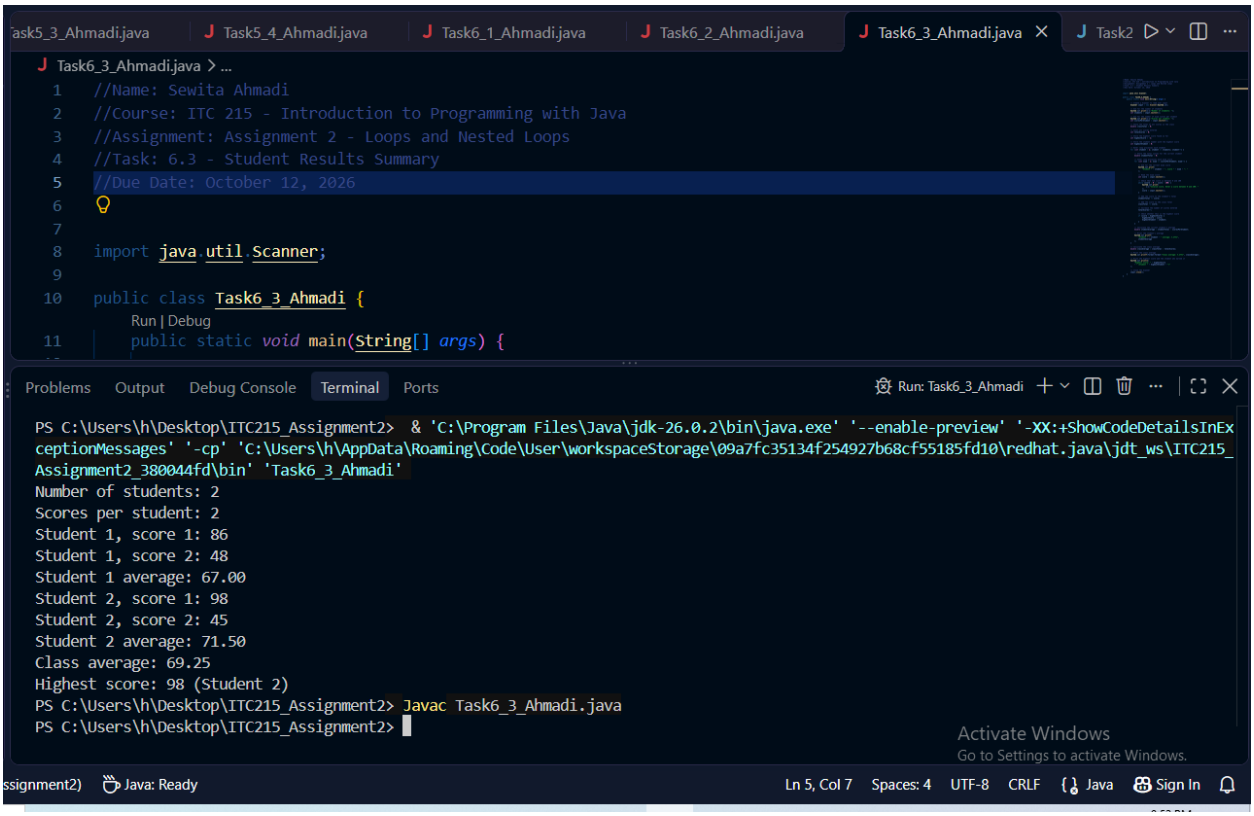


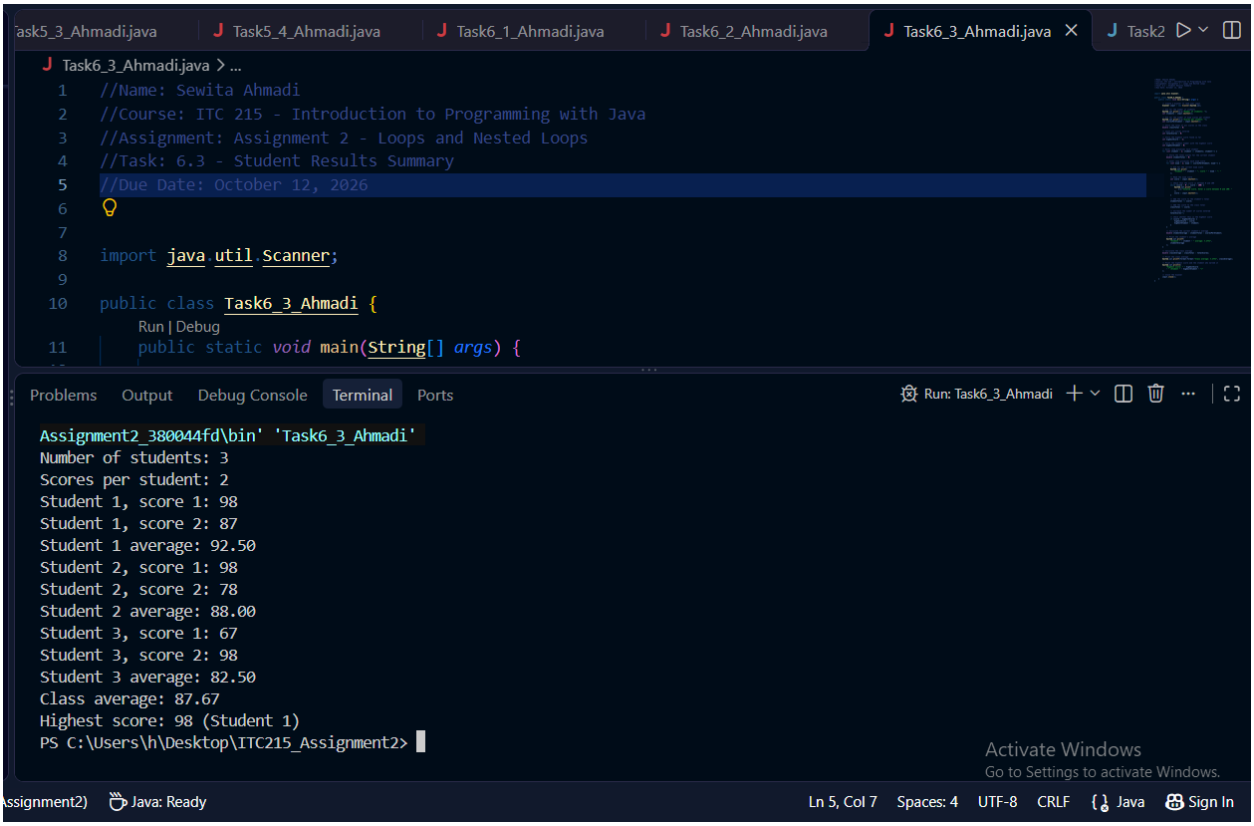
Task 6.2 Output Screenshot





Task 6.3 Output Screenshot


Task5_3_Ahmadi.java
Task5_4_Ahmadi.java
Task6_1_Ahmadi.java
Task6_2_Ahmadi.java
Task6_3_Ahmadi.java
Task2
Task6_3_Ahmadi.java > ...
1 //Name: Sewita Ahmadi
2 //Course: ITC 215 - Introduction to Programming with Java
3 //Assignment: Assignment 2 - Loops and Nested Loops
4 //Task: 6.3 - Student Results Summary
5 //Due Date: October 12, 2026
8 import java.util.Scanner;
10 public class Task6_3_Ahmadi {
Run | Debug
11 public static void main(String[] args) {
Problems
Output
Debug Console
Terminal
Ports
Run: Task6_3_Ahmadi
PS C:\Users\h\Desktop\ITC215_Assignment2> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\h\AppData\Roaming\Code\User\workspaceStorage\09a7fc35134f254927b68cf55185fd10\redhat.java\jdt_ws\ITC215_Assignment2_380044fd\bin' 'Task6_3_Ahmadi'
Number of students: 2
Scores per student: 2
Student 1, score 1: 86
Student 1, score 2: 48
Student 1 average: 67.00
Student 2, score 1: 98
Student 2, score 2: 45
Student 2 average: 71.50
Class average: 69.25
Highest score: 98 (Student 2)
PS C:\Users\h\Desktop\ITC215_Assignment2> Javac Task6_3_Ahmadi.java
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
ssignment2)
Java: Ready
Ln 5, Col 7
Spaces: 4
UTF-8
CRLF
Java
Sign In


Task5_3_Ahmadi.java
Task5_4_Ahmadi.java
Task6_1_Ahmadi.java
Task6_2_Ahmadi.java
Task6_3_Ahmadi.java
Task2
Task6_3_Ahmadi.java > ...
1 //Name: Sewita Ahmadi
2 //Course: ITC 215 - Introduction to Programming with Java
3 //Assignment: Assignment 2 - Loops and Nested Loops
4 //Task: 6.3 - Student Results Summary
5 //Due Date: October 12, 2026
8 import java.util.Scanner;
10 public class Task6_3_Ahmadi {
Run | Debug
11 public static void main(String[] args) {
Problems
Output
Debug Console
Terminal
Ports
Run: Task6_3_Ahmadi
Assignment2_380044fd\bin' 'Task6_3_Ahmadi'
Number of students: 3
Scores per student: 2
Student 1, score 1: 98
Student 1, score 2: 87
Student 1 average: 92.50
Student 2, score 1: 98
Student 2, score 2: 78
Student 2 average: 88.00
Student 3, score 1: 67
Student 3, score 2: 98
Student 3 average: 82.50
Class average: 87.67
Highest score: 98 (Student 1)
PS C:\Users\h\Desktop\ITC215_Assignment2>
Activate Windows
Go to Settings to activate Windows.
ssignment2)
Java: Ready
Ln 5, Col 7
Spaces: 4
UTF-8
CRLF
Java
Sign In